



AVP1000/AVP2000/AVP2000T

Data Sheet

BlueRiver AVP1000

Key Features

- Enables switching and extension of audio, video and control signals over standard Ethernet networks
- Compatible with full range of formats, including 4K60 RGB/4:4:4, HDR, 3D and broadcast-friendly 4K60 10-bit 4:2:2
- HDMI audio embedding, de-embedding and breakaway
- Independent switching with zero latency for all signal types
- Supports 10GBaseT copper infrastructure over category cabling
- Optical fiber transmission when paired with standard SFP+ modules
- Single device for transmitter (Tx) or receiver (Rx) products
- Combine with AVP2000 or AVP2000T for fast switching and video wall applications
- Point-to-point extension of HDMI, audio, USB, Gigabit Ethernet and control signals

BlueRiver AVP2000

Key Features

- All the features of the AVP1000 plus a powerful AV processing engine
- Broadcast quality scaler, programmable to scale up or down any resolutions between VGA and 4K
- Synchronized video wall processing with bezel correction
- Compositing engine for combining multiple sources into a single display with user-defined layouts for multi-view, tiling and picture-in-picture applications
- Audio down-mixing engine from multi-channel PCM audio to stereo
- Colour space conversion and frame rate conversion
- Supports 10G Ethernet over category cabling and optical fiber

BlueRiver AVP2000T

Key Features

- All features of the AVP2000 with simultaneous video transmit and receive functionality
- Latency-free broadcast quality scaler, programmable to scale up or down any resolutions between VGA and 4K
- Independent video scaling on both transmit and receive paths

Description

Bringing the disruptive power of Ethernet to traditional AV signal distribution, the BlueRiver chipset delivers a single programmable system on chip (SoC), which can be used to replace traditional AV/KVM extenders, matrix switchers, video wall controllers and windowing processors, with a simple network of transmitters, receivers and off the shelf Ethernet switches.

- 4K60 4:4:4 with uncompressed image quality
- Multi-channel audio, Gigabit Ethernet, RS-232, USB and IR
- Zero latency extension and switching
- Independent switching for all signal types
- Support for fiber and category cabling
- Power over Ethernet (PoE) compatible
- Scaling, compositing and image splitting
- Support for bidirectional video
- Pb-Free, Halogen Free, RoHS / WEEE compliant

All BlueRiver series chipsets are not only interoperable with each other, but also interoperable with the existing in-stalled base of SDVoE products. Sharing a common SDVoE API which intelligently enables features based on end-point capability, the AVP2000 or AVP2000T can extend the capability of the AVP1000. For example, an AVP1000 transmitter, which does not support scaling, may send its signal to an AVP2000 receiver, which can display the AVP1000 video signal with scaling, seamless switching or as a video wall.

From device discovery to runtime control, the SDVoE API makes it simple to rapidly develop and deploy products of any complexity by providing the interface for easy setup, control and monitoring of the device capabilities such as routing, scaling, video wall processing, image compositing, down-mixing and much more. The API provides interoperability between SDVoE products from a broad range of manufacturers. In point-to-point AV extender applications, the SDVoE API is not required.

With a dedicated 1GbE interface combined with an integrated 10G to 1G switch, each BlueRiver enabled endpoint can provide a 1G data port, enabling true data networking within the AV network. This capability eliminates the need for redundant equipment and cabling for locations where the AV and data network overlap.

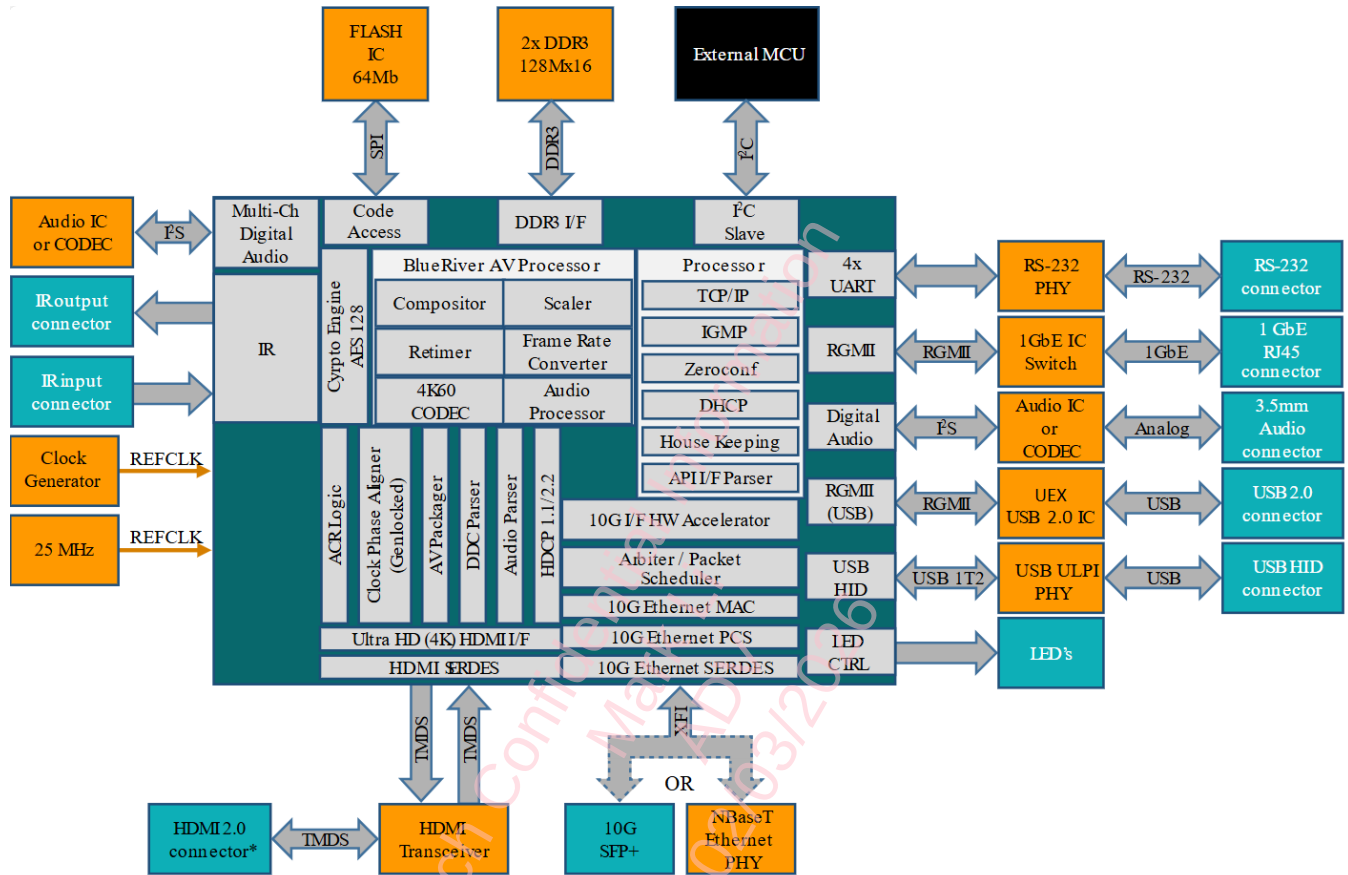
Applications

- AV/KVM extension
- AV/KVM matrix switching
- AV/KVM over IP
- Video Wall processing
- Windowing and tiling
- HDMI extenders

All of the above simultaneously on a single platform.

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Functional Block Diagrams



*AVP2000T requires two HDMI connectors

Figure A: AVP2000 and AVP2000T Block Diagram

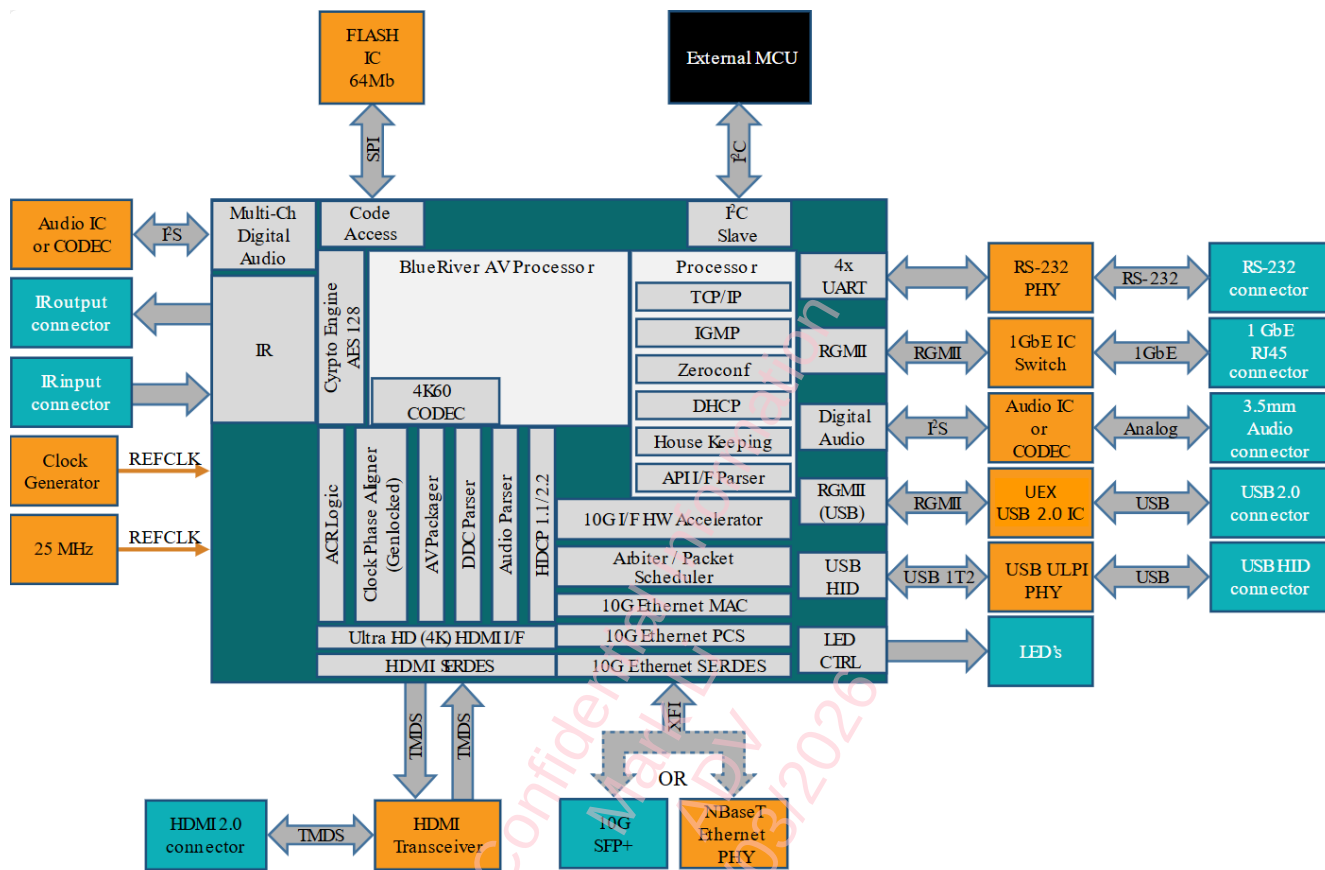


Figure B: AVP1000 Block Diagram

Comparison of BlueRiver AV Processor Device Capabilities

Table A: Comparison of BlueRiver AV Processor Device Capabilities

	AVP1000	AVP2000	AVP2000T
Video Support			
4K60 8-bit 4:4:4	✓	✓	✓
4K60 10-bit 4:2:2	✓	✓	✓
4K60 12-bit 4:2:0 (HDR)	✓	✓	✓
Transceiver Mode (Simultaneous Tx and Rx Video)	N/A	N/A	✓
Transport Medium			
Copper (10GBaseT, NBaseT)	✓	✓	✓
Multi-mode fibre (SFP+)	✓	✓	✓
Single-mode fibre (SFP+)	✓	✓	✓
AV Processing			
Zero-frame latency custom scaling to display	N/A	✓	✓
Single-frame switching	N/A	✓	✓
Video wall with bezel correction	N/A	✓	✓
Multi-view compositing	N/A	✓	✓
Audio down-mixing	✓	✓	✓
Auxiliary and Third-Party Interfaces			
1G Ethernet	✓	✓	✓
USB-HID for mouse/keyboard	✓	✓	✓
USB 2.0 with third-party chipset	✓	✓	✓
Quad RS-232, Infrared I/O, CEC, GPIO	✓	✓	✓
Single Part Number and Procurement			
Transmit or Receive Mode	✓	✓	✓
Transmit and Receive Mode	N/A	N/A	✓

Video Interfaces

- HDMI 2.0 supporting all resolutions up to 594MHz
- 4K60 RGB and 4:4:4 8-bit
- 4K60 4:2:2 10-bit for broadcast and medical applications
- 4K60 4:2:0 10-bit and 12-bit HDR

Audio Interfaces

- Integrated HDMI audio with 2-channel I2S in or out with 8-channel I2S in and out
- Supporting all HDMI 2.0 formats including multi-channel PCM, Dolby True-HD and DTS-HD master audio
- HDMI audio embedding, de-embedding & breakaway
- Each audio type independently routable
- Seamless Dante™ audio integration using available I2S and RGMII interfaces

Transport Performance

- End-to-end latency under 100 microseconds
- Video is uncompressed, except when the signal exceeds the 10GbE pipe. Streams affected include:
 - 4K60 8-bit RGB and YCbCr 4:4:4
 - 4K60 10-bit YCbCr 4:2:2
 - 4K60 12-bit YCbCr 4:2:2
 - 4K60 12-bit YCbCr 4:2:0
 - 4K30 12-bit RGB and YCbCr 4:4:4
- Ultra-lightweight compression with 1.4-to-1 compression ratio and 100% artefact-free

Transmission Interface

- 10 Gigabit Ethernet via XFI, IEEE 802.3, and 2.5/5 Gigabit interface via IEEE 802.3bz
- Compatible with 10GBaseT or NBaseT copper PHY or SFP+ fibre modules
- Power over Ethernet (PoE) support with compatible PHY devices

Peripheral Interfaces

- Quad RS-232, infrared I/O, Gigabit Ethernet, USB-HID and GPIO
- USB 2.0 full device support at 480Mb/s with third-party chipset
- Software insertion of all control signals via API

Transmission Distance

- Up to 100m over Cat-6a copper category cable when paired with 10GBaseT compliant PHY
- Fiber reach compliant with IEEE 802.3 when paired with SFP+ module:
 - SR: 300m OM3 / 400m OM4 multimode fiber
 - LR: 10km single mode fiber
 - ER: 40km single mode fiber
 - ZR: 80km single mode fiber

BlueRiver AV Processor

- Broadcast quality upscaling and downscaling
- (4K <-> VGA)
- Colour space, chroma sampling, and frame rate conversion
- Multi-view compositing up to 16x16 tiles
- Video wall with synchronized output and bezel correction up to 64 displays
- Picture-in-picture, picture-and-picture
- Multi-channel PCM audio down-mixing

General

- Power: 2.6W Typical
- Package: 576-BGA (25mm x 25mm)

Revision History

Version	ECO	PCN	Date	Changes and/or Modifications
9	074865	—	March 2025	Use Semtech UEX 2.0 Extender for RGMII USB 2.0 interface.
8	072792	—	September 2024	Added missing 1.8V support for T12 VDD_PHY10G in Table 1-1: Ball Descriptions. Added AVDD09/AVDD18 for HDMIRX/HDMITX in Table 2-3: DC Electrical Characteristics. Removed the duplicated typo of AVDD18_PLL in Table 2-4: Power Supply Maximum Noise Levels. Added section 2.2.2 Power Supply Sequencing. Added Figure 3-18: Quad I2S Audio Interface Demonstrating I2S Mode. Corrected data sampled on clock rising edge in 3.4.4 Quad I2S Interface and 3.4.5 Single I2S Interface. Added information about marking shown in Figure 5-3: Marking Diagram.
7	064436	000877	January 2023	Updates: Comparison of BlueRiver AV Processor Device Capabilities, Table 1-1: Ball Descriptions (balls C17, D16, G3, G4, H4, J2, K5, N22, P5, R2, R3, R5, T1, T2); update operating temperature in Table 2-1: Absolute Maximum Ratings from -20°C ambient minimum to +85°C junction, to -20°C junction to +100°C junction (-20°C < TJ < +100°C); add note to Table 2-2: Power; device functions in section 3.2 BlueRiver AV Processor (AVP2000 and AVP2000T) and section 3.2.2.5 Video Cropping and Bezel Correction; audio capabilities in section 3.2.3.2 Audio Embedding and De-Embedding; Single I ² S capabilities in section 3.4.5; applicable devices in 3.4.6 XFI Interface; general updates to BlueRiver device nomenclature.
6	060801	—	April 2022	Updates: Transmission Distance, Table A: Comparison of BlueRiver AV Processor Device Capabilities, Table 1-1: Ball Descriptions (balls C17, G1, H3, AC12, AD12), Table 2-5: AC Electrical Characteristics, BlueRiver Compression, Figure 3-7: RGMII Tx Delay Block Diagram and Figure 3-8: Delay on RGMII Rx.
5	057090	—	July 2021	Added section 2.2.1 Power Supply Noise, updated section 3.4.3 USB HID Interface, added Figure 3-9: USB HID Devices and Endpoints, clarifications made to Table 1-1: Ball Descriptions.
4	054073	—	October 2020	Updated Figure A: AVP2000 and AVP2000T Block Diagram, Figure B: AVP1000 Block Diagram and section 1 Ball Out.
3	052056	—	August 2020	Updates to ball descriptions to better clarify how balls should be connected when unused. Originally combined supply current for the 10G PHY, 1G RGMII and Icron RGMII supplies now broken out as individual supply currents.
2	050408	—	March 2020	Updates to Peripheral Interfaces, 1 Ball Out, Table 1-1: Ball Descriptions, Table 2-2: Power, Figure 3-1: BlueRiver Components and Interfaces, 3.2.4 Adaptive Clock Recovery (ACR), Section 3.4.1, 3.4.2 ILOCAL/REMOTE Modes.
1	048329	—	November 2019	Data sheet status changes to Preliminary. Updates to Ball Out, Electrical Characteristics, Detailed Description, and Package and Ordering Information.
0	044763	—	April 2019	New document.

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1 Ball Out

1.1 AVP2000 and AVP2000T Ball Assignment

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		
A	VSS	ICRON_USB_MODE	ICRON_PAIR	ICRON_ACTIVITY	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_TX_DATA_2	ICRON_CLK_125	VSS	HDMI_RX_TMDS_CLK_P	HDMI_RX_TMDS_CLK_N	VSS	VSS	HDMI_RX_TMDS_D0_P	HDMI_RX_TMDS_D0_N	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	HDMI_TX_TMDS_D1_P	HDMI_TX_TMDS_D1_N	VSS	VSS	A
B	VSS	ICRON_REX_LEX_N	ICRON_LINK	ICRON_VBUS_EN	VSS	ICRON_TX_DATA_3	ICRON_ETH_RST_N	VSS	VSS	VSS	HDMI_RX_TMDS_D0_P	HDMI_RX_TMDS_D0_N	VSS	VSS	HDMI_RX_TMDS_D1_P	HDMI_RX_TMDS_D1_N	VSS	VSS	HDMI_TX_TMDS_D1_P	HDMI_TX_TMDS_D1_N	VSS	VSS	HDMI_TX_TMDS_D2_P	HDMI_TX_TMDS_D2_N	VSS	B
C	ICRON_RST	ICRON_BOOT_SEL0	ICRON_UART_TX	ICRON_PHYSEL0	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_TX_DATA_2	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_TX_DATA_2	HDMI_RX_HFCHM1_RX_REST	VSS	VSS	HDMI_TX_RX_TX_CLK_P	HDMI_TX_RX_TX_CLK_N	VSS	VSS	HDMI_TX_CEC	VSS	VSS	VSS	HDMI_TX_TX_HFCHM1_TX_HPD	VSS	ICRON_DQ_0_P	ICRON_DQ_0_N	C
D	GPIO_1	GPIO_11	VSS	ICRON_UART_RX	ICRON_HDMI_TX_DATA_3	ICRON_HDMI_TX_DATA_1	ICRON_HDMI_TX_DATA_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	HDMI_RX_TX_TX_CLK_P	HDMI_RX_TX_TX_CLK_N	VSS	VSS	HDMI_TX_TX_HFCHM1_TX_HPD	VSS	VSS	ICRON_DQ_1_P	ICRON_DQ_1_N	ICRON_DQ_2_P	D
E	GPIO_4	GPIO_9	GPIO_7	GPIO_0	GPIO_8	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_TX_DATA_2	VSS	VSS	VSS	VSS	VSS	HDMI_RX_TX_TX_CLK_P	HDMI_RX_TX_TX_CLK_N	VSS	VSS	HDMI_TX_TX_HFCHM1_TX_HPD	VSS	VSS	VSS	VSS	ICRON_DQ_3_P	ICRON_DQ_3_N	VSS	E
F	RSVD_6	GPIO_5	VSS	GPIO_6	GPIO_10	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_4_P	ICRON_DQ_4_N	ICRON_DQ_5_P	F
G	CLK_MNG_FRC_FB	RSVD_4	SINGLE_125_SPI_SCLK	SINGLE_125_SPI_CS0	GPIO_3	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_6_P	ICRON_DQ_6_N	ICRON_DQ_7_P	G
H	RSVD_5	VSS	CLK_MNG_FRC_FB	SINGLE_125_SPI_CS0	GPIO_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_8_P	ICRON_DQ_8_N	ICRON_DQ_9_P	H
J	LED_USB_PRESENT	LED_VIDEO_OUT_STABLE	LED_10G_TX_ACTIVITY	VSS	LED_USB_ACTIVITY	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_11_P	ICRON_DQ_11_N	ICRON_DQ_12_P	J
K	LED_SYSTEM_STATUS	VSS	LED_10G_TX_ACTIVITY	LED_USB_ACTIVITY	LED_VIDEO_OUT_STABLE	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_14_P	ICRON_DQ_14_N	ICRON_DQ_15_P	K
L	QUAD_125_RX_OUT_0	QUAD_125_RX_OUT_1	QUAD_125_RX_OUT_2	QUAD_125_RX_OUT_3	QUAD_125_RX_OUT_4	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_17_P	ICRON_DQ_17_N	ICRON_DQ_18_P	L
M	QUAD_125_RX_OUT_7	QUAD_125_RX_OUT_8	VSS	QUAD_125_RX_OUT_9	QUAD_125_RX_OUT_10	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_20_P	ICRON_DQ_20_N	ICRON_DQ_21_P	M
N	QUAD_125_RX_OUT_11	QUAD_125_RX_OUT_12	QUAD_125_RX_OUT_13	QUAD_125_RX_OUT_14	QUAD_125_RX_OUT_15	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_23_P	ICRON_DQ_23_N	ICRON_DQ_24_P	N
P	REQ_SDA	VSS	QUAD_125_RX_OUT_16	QUAD_125_RX_OUT_17	SFP_SDA	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_26_P	ICRON_DQ_26_N	ICRON_DQ_27_P	P
R	REQ_SCL	SFP_TX_DISABLE	SFP_RATE_SEL	VSS	SFP_SCL	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_29_P	ICRON_DQ_29_N	ICRON_DQ_30_P	R
T	SFP_L0BS	SFP_PRESENT	RSVD_1	RSVD_2	SINGLE_125_SPI_CS0	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_32_P	ICRON_DQ_32_N	ICRON_DQ_33_P	T
U	SINGLE_125_SPI_CS0	SINGLE_125_SPI_CS1	VSS	SINGLE_125_SPI_CS2	RSVD_0	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_35_P	ICRON_DQ_35_N	ICRON_DQ_36_P	U
V	SINGLE_125_SPI_CS1	SINGLE_125_SPI_CS2	ULPI_DATA_0	ULPI_DATA_1	ULPI_DATA_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_38_P	ICRON_DQ_38_N	ICRON_DQ_39_P	V
W	ULPI_STP	ULPI_CLK	ULPI_DATA_3	ULPI_DATA_4	ULPI_DATA_5	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_41_P	ICRON_DQ_41_N	ICRON_DQ_42_P	W
Y	VSS	VSS	ULPI_DATA_6	ULPI_DATA_7	VSS	ULPI_DATA_8	ULPI_DATA_9	ULPI_DATA_10	ULPI_DATA_11	ULPI_DATA_12	ULPI_DATA_13	ULPI_DATA_14	ULPI_DATA_15	ULPI_DATA_16	ULPI_DATA_17	ULPI_DATA_18	ULPI_DATA_19	ULPI_DATA_20	ULPI_DATA_21	ULPI_DATA_22	ULPI_DATA_23	ULPI_DATA_24	ULPI_DATA_25	ULPI_DATA_26	Y	
AA	RSVD_20	RSVD_19	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_44_P	ICRON_DQ_44_N	ICRON_DQ_45_P	AA
AB	VSS	VSS	ULPI_DATA_27	ULPI_DATA_28	ULPI_DATA_29	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_47_P	ICRON_DQ_47_N	ICRON_DQ_48_P	AB
AC	ULPI_DATA_30	ULPI_DATA_31	VSS	VSS	ULPI_DATA_32	ULPI_DATA_33	ULPI_DATA_34	ULPI_DATA_35	ULPI_DATA_36	ULPI_DATA_37	ULPI_DATA_38	ULPI_DATA_39	ULPI_DATA_40	ULPI_DATA_41	ULPI_DATA_42	ULPI_DATA_43	ULPI_DATA_44	ULPI_DATA_45	ULPI_DATA_46	ULPI_DATA_47	ULPI_DATA_48	ULPI_DATA_49	ULPI_DATA_50	ULPI_DATA_51	AC	
AD	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	ICRON_DQ_52_P	ICRON_DQ_52_N	ICRON_DQ_53_P	AD

Figure 1-1: AVP2000/AVP2000T Ball Assignment

	1	2	3	4	5	6	7	8	9	10	11	12
A	VSS	ICRON_USB_MODE	ICRON_PAIR_N	ICRON_ACTIVITY	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_CLK_125	VSS	HDMI_RX_TMDS_CLK_N	HDMI_RX_TMDS_CLK_P	VSS	VSS
B	VSS	ICRON_REX_LEX_N	ICRON_LINK	ICRON_VBUS_EN	VSS	ICRON_TX_DATA_2	ICRON_ETH_RST_N	VSS	VSS	VSS	HDMI_RX_TMDS_D0_I	HDMI_RX_TMDS_D0_P
C	ICRON_RST	ICRON_BOOT_SEL0	ICRON_UART_TX	ICRON_PHY2LEN	ICRON_TX_DATA_0	ICRON_RX_DATA_0	ICRON_TX_CLK	ICRON_MDC	HDMI_RX_HPD	HDMI_RX_REXT	VSS	VSS
D	GPIO_1	GPIO_11	VSS	ICRON_UART_RX	ICRON_HOST	ICRON_RX_CLK	ICRON_RX_DATA_3	ICRON_RX_DATA_2	VSS	VSS	VSS	VSS
E	GPIO_4	GPIO_9	GPIO_7	GPIO_0	GPIO_8	ICRON_TX_CLK	ICRON_RX_CLK	ICRON_RX_DATA_1	ICRON_MDC	VSS	VSS	VSS
F	RSVD_6	GPIO_5	VSS	GPIO_6	GPIO_10	VSS	VSS	VSS	VSS	VSS	VSS	VSS
G	CLK_MNG_FINC_FB	RSVD_4	SINGLE_I2S_SPI_SCLK	SINGLE_I2S_SPI_CSB	GPIO_3	VSS	VSS	VSS	VSS	VSS	VSS	VSS
H	RSVD_5	VSS	CLK_MNG_FDEC_FB	SINGLE_I2S_SPI_SDIN	GPIO_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS
J	LED_USB_PRESENT	LED_VIDEO_OUT_STABLE	LED_10G_TX_ACTIVITY	VSS	LED_USB_HI_ACTIVITY	VSS	VSS	VSS	VSS	VSS	AVDD18_HDM	AVDD33_HDM
K	LED_SYSTEM_STATUS	VSS	LED_10G_RX_ACTIVITY	LED_USB_ACTIVITY	LED_VIDEO_IN_STABLE	VSS	VSS	VSS	VSS	VSS	AVDD09_HDM RX	VSS
L	QUAD_I2S_SOUT_0	IR_DATA_IN	IR_DATA_OUT	QUAD_I2S_SOUT_3	QUAD_I2S_WS_OUT	VSS	VSS	VSS	VSS	VSS	VDD_ICRON	VDD09
M	QUAD_I2S_SOUT_2	QUAD_I2S_SCK_IN	VSS	QUAD_I2S_SOUT_IN_2	QUAD_I2S_SOUT_1	VSS	VSS	VSS	VSS	VSS	VDD33	VSS

Figure 1-2: AVP2000/AVP2000T Ball Assignment Quadrant 1

13	14	15	16	17	18	19	20	21	22	23	24	
HDMI_RX_TMDS_D1_N	HDMI_RX_TMDS_D1_P	VSS	VSS	HDMI_TX_TMDS_D2_P	HDMI_TX_TMDS_D2_N	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	A
VSS	VSS	HDMI_RX_TMDS_D2_N	HDMI_RX_TMDS_D2_P	VSS	VSS	HDMI_TX_TMDS_D1_P	HDMI_TX_TMDS_D1_N	VSS	VSS	HDMI_TX_TMDS_CLK_P	HDMI_TX_TMDS_CLK_N	B
HDMI_TX_REC_CLK_N	HDMI_TX_REC_CLK_P	VSS	VSS	HDMI_TX_CEC	VSS	VSS	VSS	HDMI_TX_5V	HDMI_TX_HPD	VSS	DDR3_DQ_21	C
VSS	VSS	HDMI_RX_5V	HDMI_RX_CEC	VSS	VSS	VSS	HDMI_TX_REXT	VSS	VSS	DDR3_DQ_19	DDR3_DM_2	D
HDMI_RX_DDC_SCL	HDMI_RX_DDC_SDA	VSS	HDMI_TX_DDC_SCL	HDMI_TX_DDC_SDA	VSS	VSS	VSS	VSS	DDR3_DQ_17	DDR3_DQ_20	VSS	E
VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	DDR3_DQ_23	DDR3_DQ_26	DDR3_DQS_2_P	DDR3_DQS_2_N	F
VSS	VSS	VSS	VSS	VSS	VSS	DDR3_DQ_22	DDR3_DQ_18	DDR3_DQ_28	VSS	DDR3_DQ_31	VSS	G
VSS	VSS	VSS	RSVD_16	VSS	VSS	DDR3_DQ_16	DDR3_DQ_29	DDR3_DQ_25	DDR3_VREF_2	DDR3_DQS_3_N	DDR3_DQS_3_P	H
VDD33	AVDD18_LVDS	AVDD18_HDMI_TX	VSS	VSS	VSS	DDR3_DQ_30	VSS	DDR3_ADDR_6	DDR3_DQ_24	VSS	DDR3_ADDR_11	J
AVDD09_HDMI_TX	VSS	VDD18_DDR	VSS	VSS	VSS	DDR3_DM_5	DDR3_DQ_27	DDR3_RESET	DDR3_ADDR_2	DDR3_ADDR_3	DDR3_ADDR_4	K
VSS	VDD09_DDR	VDD_DDR	VSS	VSS	VSS	DDR3_VREF	VSS	DDR3_ADDR_2	DDR3_BA_2	DDR3_ZQ_05_P	VSS	L
VDD09_DDR	VSS	VDD_DDR	VSS	VSS	VSS	DDR3_ADDR_8	DDR3_ZQ_05_N	DDR3_BA_1	VSS	DDR3_ADDR_0	DDR3_CKE	M

Figure 1-3: AVP2000/AVP2000T Ball Assignment Quadrant 2

N	QUAD_I2S_SD_IN_1	QUAD_I2S_SD_IN_1	QUAD_I2S_MCLK_OUT	QUAD_I2S_WS_IN	QUAD_I2S_MCLK_IN	VSS	VSS	VSS	VSS	VSS	VDD33	VDD09
P	REG_SDA	VSS	QUAD_I2S_SD_IN_0	QUAD_I2S_SD_IN_3	SFP_SDA	VSS	VSS	VSS	VSS	AVDD18_PLL	AVDD09_CML	VSS
R	REG_SCL	SFP_TX_DISABLE	SFP_RATE_SEL	VSS	SFP_SCL	VSS	VSS	VSS	VSS	AVDD09_XFI	AVDD18_XFI	VDD_PHY1G
T	SFP_LOSS	SFP_PRESENT	N_RSVD_1	RSVD_2	SINGLE_I2S_DIR	VSS	VSS	VSS	VSS	VSS	VSS	VDD_PHY10G
U	SINGLE_I2S_LRC	SINGLE_I2S_BCLK	VSS	SINGLE_I2S_SD_OUT	RSVD_0	VSS	VSS	VSS	VSS	VSS	VSS	VSS
V	SINGLE_I2S_SD_IN	SINGLE_I2S_MCLK	ULPI_DATA_0	ULPI_DATA_1	ULPI_DATA_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS
W	ULPI_STP	ULPI_CLOCK	ULPI_NXT	ULPI_DATA_3	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS
Y	VSS	VSS	ULPI_DIR	ULPI_RESET_N	VSS	XFI_TX0_POLARITY	ULPI_DATA_4	PHY_10G_IN_N	PHY_1G_TX_DATA_2	PHY_1G_RX_DATA_1	PHY_1G_RX_DATA_2	UART_2_RX
AA	RSVD_20	RSVD_19	VSS	VSS	VSS	XFI_RX0_POLARITY	PHY_10G_MCLK	ULPI_DATA_5	PHY_1G_TX_DATA_3	VSS	PHY_1G_RX_CLK	UART_4_RX
AB	VSS	VSS	XFI_RX0_N	XFI_RX0_P	VSS	ULPI_DATA_6	VSS	PHY_1G_TX_DATA_1	VSS	PHY_1G_RX_CTL	UART_3_RX	UART_4_TX
AC	XFI_TX0_N	XFI_TX0_P	VSS	VSS	ULPI_DATA_7	PHY_10G_RST_OUT_N	PHY_10G_TX_EN	PHY_1G_TX_DATA_0	PHY_1G_TX_CLK	PHY_1G_RX_DATA_0	UART_3_TX	UART_1_RX
AD	VSS	VSS	VSS	XFI_REXT	ULPI_CS_N	VSS	PHY_10G_MDIO	VSS	PHY_1G_TX_CTL	VSS	PHY_1G_RX_DATA_3	UART_1_TX
	1	2	3	4	5	6	7	8	9	10	11	12

Figure 1-4: AVP2000/AVP2000T Ball Assignment Quadrant 3

VSS	AVDD18_DDR_PL	VDD_DDR	VSS	VSS	VSS	VSS	DDR3_CS_N	DDR3_ODT	DDR3_CAS_N	DDR3_CK_P	DDR3_CK_N	N
VDD33	VDD09_DDR	AVDD18_DDR	VSS	VSS	VSS	DDR3_ADDR_0	DDR3_RAS_N	VSS	DDR3_WE_N	VSS	VSS	P
VSS	VDD18	VSS	VSS	VSS	VSS	VSS	DDR3_ADDR_1	DDR3_ADDR_2	DDR3_ADDR_3	DDR3_BA_0	DDR3_ADDR_3	R
VSS	VSS	VDD18_JTM	VSS	VSS	VSS	DDR3_DQ_14	DDR3_ZQ_08_P	VSS	DDR3_ZQ_08_N	DDR3_ADDR_4	VSS	T
VSS	VSS	VSS	VSS	VSS	VSS	VSS	DDR3_ADDR_7	DDR3_DQ_8	DDR3_DQ_12	DDR3_DQS_1_N	DDR3_DQS_1_P	U
VSS	VSS	VSS	VSS	VSS	VSS	DDR3_DQ_7	DDR3_DQ_13	DDR3_DQ_15	VSS	DDR3_DQ_15	VSS	V
VSS	VSS	VSS	VSS	VSS	VSS	VSS	DDR3_DQ_1	DDR3_DM_1	DDR3_DQ_9	DDR3_DQ_10	DDR3_DQ_6	W
UART_2_TX	FW_FLASH_SCLK	CLK_MNG_SC	CLK_MNG_SDA	IREF	RSVD_9	VSS	VSS	DDR3_DM_0	DDR3_DQ_3	DDR3_DQ_5	VSS	Y
VSS	FW_FLASH_IO3	INT	VSS	A_SEL1	RSVD_15	VSS	VSS	DDR3_DQ_2	VSS	DDR3_DQS_0_P	DDR3_DQS_0_N	AA
FW_FLASH_CS_N	VSS	CLK_MNG_RESET_N	CLK_MNG_FINC	RST_N	RSVD_10	RSVD_14	TCK	TRST	DDR3_VREF_1	DDR3_DQ_0	DDR3_DQ_4	AE
BOOT_MODE	FW_FLASH_IO0	FW_FACTORY	CLK_MNG_FDEC	ULPI_VBUS	A_SEL0	RSVD_13	TDO	TDI	RSVD_17	RSVD_18	VSS	AC
VSS	FW_FLASH_IO1	FW_FLASH_IO2	VSS	RSVD_12	RSVD_11	RSVD_8	VSS	XTAL_25_IN	XTAL_25_OUT	TMS	VSS	AC
13	14	15	16	17	18	19	20	21	22	23	24	

Figure 1-5: AVP2000/AVP2000T Ball Assignment Quadrant 4

1.2 AVP1000 Ball Assignment

The AVP1000 differs in that there are no DDR Memory allocated balls, and these are now labelled as reserved. These balls are also contained in quadrants 2 and 4 of Figure 1-6, hence only those quadrants will be detailed separately in Figure 1-7 and Figure 1-8 below. Quadrant 1 and 3 are identical to AVP2000 & AVP2000T.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		
A	VSS	ICRON_USB_MODE	ICRON_PAIR	ICRON_ACTIVITY	ICRON_TX_DATA_3	ICRON_TX_DATA_1	ICRON_CLK_125	VSS	HDMI_RX_TMDS_CLK_N	HDMI_RX_TMDS_CLK_P	VSS	VSS	HDMI_RX_TMDS_D0_N	HDMI_RX_TMDS_D0_P	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	A	
B	VSS	ICRON_RX_LEX_N	ICRON_LINK	ICRON_VBUS_EN	VSS	ICRON_TX_DATA_2	ICRON_ETH_RST_N	VSS	VSS	VSS	HDMI_RX_TMDS_D0_N	HDMI_RX_TMDS_D0_P	VSS	VSS	HDMI_RX_TMDS_D0_N	HDMI_RX_TMDS_D0_P	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	HDMI_TX_TMDS_CLK_N	HDMI_TX_TMDS_CLK_P	B	
C	ICRON_RST	ICRON_BOOT_SEL0	ICRON_UART_TX	ICRON_PHYZEN	ICRON_TX_DATA_0	ICRON_TX_DATA_0	ICRON_TX_CLK	ICRON_MIOCE	HDMI_RX_HPD	HDMI_RX_HPD	VSS	VSS	HDMI_TX_REC_CLK_N	HDMI_TX_REC_CLK_P	VSS	VSS	HDMI_TX_CEC	VSS	VSS	VSS	VSS	HDMI_TX_SV	HDMI_TX_HPD	VSS	RSVD	C
D	GPIO_1	GPIO_11	VSS	ICRON_UART_RX	ICRON_HOST	ICRON_RX_CLK	ICRON_RX_DATA_3	ICRON_RX_DATA_2	VSS	VSS	VSS	VSS	VSS	VSS	HDMI_RX_FV	HDMI_RX_CEC	VSS	VSS	VSS	HDMI_TX_REXT	VSS	VSS	RSVD	RSVD	D	
E	GPIO_4	GPIO_9	GPIO_7	GPIO_0	GPIO_8	ICRON_TX_CLK	ICRON_RX_CLK	ICRON_TX_DATA_1	ICRON_MIOCE	VSS	VSS	VSS	HDMI_RX_DDC_SCL	HDMI_RX_DDC_SDA	VSS	HDMI_TX_DDC_SCL	HDMI_TX_DDC_SDA	VSS	VSS	VSS	VSS	RSVD	RSVD	VSS	E	
F	RSVD_6	GPIO_5	VSS	GPIO_6	GPIO_10	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	F	
G	CLK_MNG_FNC_FB	RSVD_4	SINGLE_I2S_SPI_SCLK	SINGLE_I2S_SPI_DIN	GPIO_3	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	VSS	RSVD	G	
H	RSVD_5	VSS	CLK_MNG_FOSC_FB	SINGLE_I2S_SPI_SDA	GPIO_2	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD_16	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	H	
J	LED_USB_PRESENT	LED_VIDEO_OUT_STABLE	LED_10G_TX_ACTIVITY	VSS	LED_USB_HV_ACTIVITY	VSS	VSS	VSS	VSS	VSS	AVDD18_HDMI_RX_D03	HDMI_RX_VDD03	AVDD18_HDMI_TX_VSS	VSS	VSS	VSS	VSS	VSS	RSVD	VSS	RSVD	RSVD	VSS	RSVD	J	
K	LED_SYSTEM_STATUS	VSS	LED_10G_RX_ACTIVITY	LED_USB_ACTIVITY	LED_VIDEO_HV_STABLE	VSS	VSS	VSS	VSS	VSS	AVDD09_HDMI_TX_VSS	AVDD09_HDMI_TX_VSS	VDD19_D0R	VSS	VDD19_D0R	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	K	
L	QUAD_I2S_S_OUT_0	IR_DATA_IN	IR_DATA_OUT	QUAD_I2S_S_OUT_3	QUAD_I2S_S_OUT_1	VSS	VSS	VSS	VSS	VSS	VDD09	VDD09	VSS	VDD09_D0R	VSS	VSS	VSS	VSS	RSVD	VSS	RSVD	RSVD	RSVD	VSS	L	
M	QUAD_I2S_S_OUT_2	QUAD_I2S_S_OUT_1	VSS	QUAD_I2S_S_OUT_2	QUAD_I2S_S_OUT_1	VSS	VSS	VSS	VSS	VSS	VDD09	VSS	VDD09_D0R	VSS	VDD09_D0R	VSS	VSS	VSS	RSVD	RSVD	RSVD	VSS	RSVD	RSVD	M	
N	QUAD_I2S_S_OUT_1	QUAD_I2S_S_OUT_1	QUAD_I2S_S_OUT_1	QUAD_I2S_S_OUT_1	QUAD_I2S_S_OUT_1	VSS	VSS	VSS	VSS	VSS	VDD03	VDD09	VSS	VDD09_D0R	VSS	VDD09_D0R	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	N	
P	REQ_SDA	VSS	QUAD_I2S_S_OUT_0	QUAD_I2S_S_OUT_3	SFP_SDA	VSS	VSS	VSS	VSS	VSS	AVDD18_HV	AVDD18_HV	VSS	VDD03	VDD09_D0R	VDD09_D0R	VSS	VSS	VSS	RSVD	RSVD	VSS	RSVD	VSS	P	
R	REQ_SCL	SFP_TX_DISABLE	SFP_RATE_SEL	VSS	SFP_SCL	VSS	VSS	VSS	VSS	VSS	AVDD09_HV	AVDD09_HV	VDD19_D0R	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	R	
T	SFP_LOSS	SFP_PRESENT	N_RSVD_1	RSVD_2	SINGLE_I2S_DIR	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	VSS	RSVD	RSVD	VSS	T	
U	SINGLE_I2S_LRC	SINGLE_I2S_SCLK	VSS	SINGLE_I2S_SCLK	RSVD_0	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	U	
V	SINGLE_I2S_SDA_IN	SINGLE_I2S_SCLK	ULPI_DATA0	ULPI_DATA1	ULPI_DATA2	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	VSS	RSVD	VSS	V	
W	ULPI_STP	ULPI_CLOD	ULPI_NXT	ULPI_DATA3	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	W	
Y	VSS	VSS	ULPI_DIR	ULPI_RESET	VSS	XFI_TX0_POLARITY	ULPI_DATA	PHY_10G_IN	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	UART_2_RX	UART_2_TX	FW_FLASH_SCLK	CLK_MNG_SCLK	CLK_MNG_SDA	IREF	RSVD_9	VSS	VSS	RSVD	RSVD	VSS	Y	
AA	RSVD_20	RSVD_19	VSS	VSS	VSS	XF1_RX0_POLARITY	PHY_10G_MQ	ULPI_DATA	PHY_10G_TX_DATA_3	VSS	PHY_10G_TX_DATA_1	PHY_10G_TX_DATA_3	UART_4_RX	VSS	FW_FLASH_RST_N	INT	VSS	A_SEL1	RSVD_15	VSS	VSS	RSVD	VSS	RSVD	AA	
AB	VSS	VSS	XF1_RX0_N	XF1_RX0_P	VSS	ULPI_DATA	VSS	PHY_10G_TX_DATA_1	VSS	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	UART_3_RX	UART_4_TX	FW_FLASH_CS_N	VSS	CLK_MNG_RST_N	CLK_MNG_FNC	RST_N	RSVD_10	RSVD_14	TOK	TRST	RSVD	RSVD	AB	
AC	XF1_TX0_N	XF1_TX0_P	VSS	VSS	ULPI_DATA	PHY_10G_RX_DATA_0	PHY_10G_TX_DATA_0	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	UART_3_TX	UART_1_RX	BOOT_MODE	FW_FLASH_IOR	FACTORY	CLK_MNG_FDEC	ULPI_VBUS	A_SEL0	RSVD_13	TDO	TDI	RSVD_17	RSVD_16	AC	
AD	VSS	VSS	VSS	XF1_REXT	ULPI_CS_N	VSS	PHY_10G_MQ	VSS	PHY_10G_TX_DATA_0	VSS	PHY_10G_TX_DATA_3	PHY_10G_TX_DATA_1	UART_1_TX	VSS	FW_FLASH_IOR	FW_FLASH_CS_N	VSS	RSVD_12	RSVD_11	RSVD_8	VSS	XTAL_25	XTAL_25_OUT	TMS	VSS	AD
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		

Figure 1-6: AVP1000 Ball Assignment

13	14	15	16	17	18	19	20	21	22	23	24	
HDMI_RX_TMDS_D1_N	HDMI_RX_TMDS_D1_P	VSS	VSS	HDMI_TX_TMDS_D2_P	HDMI_TX_TMDS_D2_N	VSS	VSS	HDMI_TX_TMDS_D0_P	HDMI_TX_TMDS_D0_N	VSS	VSS	A
VSS	VSS	HDMI_RX_TMDS_D2_N	HDMI_RX_TMDS_D2_P	VSS	VSS	HDMI_TX_TMDS_D1_P	HDMI_TX_TMDS_D1_N	VSS	VSS	HDMI_TX_TMDS_CLK_P	HDMI_TX_TMDS_CLK_N	B
HDMI_TX_REC_CLK_N	HDMI_TX_REC_CLK_P	VSS	VSS	HDMI_TX_CEC	VSS	VSS	VSS	HDMI_TX_5V	HDMI_TX_HPD	VSS	RSVD	C
VSS	VSS	HDMI_RX_5V	HDMI_RX_CEC	VSS	VSS	VSS	HDMI_TX_REXT	VSS	VSS	RSVD	RSVD	D
HDMI_RX_DDC_SCL	HDMI_RX_DDC_SDA	VSS	HDMI_TX_DDC_SCL	HDMI_TX_DDC_SDA	VSS	VSS	VSS	VSS	RSVD	RSVD	VSS	E
VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	F
VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	VSS	RSVD	VSS	G
VSS	VSS	VSS	RSVD_16	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	H
VDD33	AVDD18_LVDS	AVDD18_HDMI_TX	VSS	VSS	VSS	RSVD	VSS	RSVD	RSVD	VSS	RSVD	J
AVDD09_HDMI_TX	VSS	VDD18_DDR	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	K
VSS	VDD09_DDR	VDD_DDR	VSS	VSS	VSS	RSVD	VSS	RSVD	RSVD	RSVD	VSS	L
VDD09_DDR	VSS	VDD_DDR	VSS	VSS	VSS	RSVD	RSVD	RSVD	VSS	RSVD	RSVD	M

Figure 1-7: AVP1000 Ball Assignment Quadrant 2

VSS	AVDD18_DDR_PL	VDD_DDR	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	N
VDD33	VDD09_DDR	VDD18_DDR	VSS	VSS	VSS	RSVD	RSVD	VSS	RSVD	VSS	VSS	P
VSS	VDD18	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	R
VSS	VSS	VDD18_JTM	VSS	VSS	VSS	RSVD	RSVD	VSS	RSVD	RSVD	VSS	T
VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	U
VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	VSS	RSVD	VSS	V
VSS	VSS	VSS	VSS	VSS	VSS	VSS	RSVD	RSVD	RSVD	RSVD	RSVD	W
UART_2_TX	FW_FLASH_SCLK	CLK_MNG_SC	CLK_MNG_SDA	IREF	RSVD_9	VSS	VSS	RSVD	RSVD	RSVD	VSS	Y
VSS	FW_FLASH_IO3	INT	VSS	A_SEL1	RSVD_15	VSS	VSS	RSVD	VSS	RSVD	RSVD	AA
FW_FLASH_CS_N	VSS	CLK_MNG_RESET_N	CLK_MNG_FINC	RST_N	RSVD_10	RSVD_14	TCK	TRST	RSVD	RSVD	RSVD	AE
BOOT_MODE	FW_FLASH_IO0	FACTORY	CLK_MNG_FDEC	ULPI_VBUS	A_SEL0	RSVD_13	TDO	TDI	RSVD_17	RSVD_18	VSS	AC
VSS	FW_FLASH_IO1	FW_FLASH_IO2	VSS	RSVD_12	RSVD_11	RSVD_8	VSS	XTAL_25_IN	XTAL_25_OUT	TMS	VSS	AC
13	14	15	16	17	18	19	20	21	22	23	24	

Figure 1-8: AVP1000 Ball Assignment Quadrant 4

1.3 AVP1000/AVP2000/AVP2000T Ball Descriptions

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ADV
02/03/2026

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
A1, A8, A11, A12, A15, A16, A19, A20, A23, A24, B1, B5, B8, B9, B10, B13, B14, B17, B18, B21, B22, C11, C12, C15, C16, C18, C19, C20, C23, D3, D9, D10, D11, D12, D13, D14, D17, D18, D19, D21, D22, E10, E11, E12, E15, E18, E19, E20, E21, E24, F3, F6-F20, G6-G18, G22, G24, H2, H6-H15, H17, H18, J4, J6-J10, J16-J18, J20, J23, K2, K6-K10, K12, K14, K16-K18, L6-L10, L13, L16-L18, L20, L24, M3, M6-M10, M12, M14, M16-M18, M22, N6-N10, N13, N16-N19, P2, P6-P9, P12, P16-P18, P21, P23, P24, R4, R6-R9, R13, R15-R19, T6-T11, T13, T14, T16-T18, T21, T24, U3, U6-U19, V6-V18, V22, V24, W5-W19, Y1, Y2, Y5, Y19, Y20, Y24, AA3-AA5, AA10, AA13, AA16, AA19, AA20, AA22, AB1, AB2, AB5, AB7, AB9, AB14, AC3, AC4, AC24, AD1-AD3, AD6, AD8, AD10, AD13, AD16, AD20, AD24	VSS	Power	Device ground. Connect to GND.
A2	UEX_USB_MODE	Digital Output	Semtech UEX USB 2.0 extender CFG. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
A3	UEX_PAIR_N	Digital Output	Semtech UEX USB 2.0 extender CFG. Active low. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
A4	UEX_ACTIVITY	Digital Input	Semtech UEX USB 2.0 extender STATUS. This ball has an internal weak pull-down resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
A5, B6, A6, C5	UEX_TX_DATA_[3:0]	Digital Output	Semtech UEX USB 2.0 extender RGMII Transmit Data. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
A7	UEX_CLK_125	Digital Output	Semtech UEX USB 2.0 Extender RGMII Reference CLK. Operates at 125MHz. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
A10, A9	HDMI_RX_TMDS_CLK_P, HDMI_RX_TMDS_CLK_N	Analog/PHY Input	HDMI Rx TMDS Clock differential input. When the HDMI RX interface is not used, this ball should be left open (no connect).
A14, A13	HDMI_RX_TMDS_D1_P, HDMI_RX_TMDS_D1_N	Analog/PHY Input	HDMI Rx TMDS Data 1 differential input. When the HDMI RX interface is not used, this ball should be left open (no connect).
A17, A18	HDMI_TX_TMDS_D2_P, HDMI_TX_TMDS_D2_N	Analog/PHY Output	HDMI Tx TMDS Data 2 differential output. When the HDMI TX interface is not used, this ball should be left open (no connect).

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
A21, A22	HDMI_TX_TMDS_D0_P, HDMI_TX_TMDS_D0_N	Analog/PHY Output	HDMI Tx TMDS Data 0 differential output. When the HDMI TX interface is not used, this ball should be left open (no connect).
B2	UEX_REX_LEX_N	Digital Output	Semtech UEX USB 2.0 extender CFG. Active low. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
B3	UEX_LINK	Digital Input	Semtech UEX USB 2.0 extender STATUS. This ball has an internal weak pull-down resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
B4	UEX_VBUS_EN	Digital Input	Semtech UEX USB 2.0 extender CFG. This ball has an internal weak pull-down resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
B7	UEX_ETH_RST_N	Digital Input	Semtech UEX USB 2.0 extender CFG. Active low. This ball has an internal weak pull-down resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
B12, B11	HDMI_RX_TMDS_D0_P, HDMI_RX_TMDS_D0_N	Analog/PHY Input	HDMI Rx TMDS Data 0 differential input. When the HDMI RX interface is not used, this ball should be left open (no connect).
B16, B15	HDMI_RX_TMDS_D2_P, HDMI_RX_TMDS_D2_N	Analog/PHY Input	HDMI Rx TMDS Data 2 differential input. When the HDMI RX interface is not used, this ball should be left open (no connect).
B19, B20	HDMI_TX_TMDS_D1_P, HDMI_TX_TMDS_D1_N	Analog/PHY Output	HDMI Tx TMDS Data 1 differential output. When the HDMI TX interface is not used, this ball should be left open (no connect).
B23, B24	HDMI_TX_TMDS_CLK_P, HDMI_TX_TMDS_CLK_N	Analog/PHY Output	HDMI Tx TMDS Clock differential output. When the HDMI TX interface is not used, this ball should be left open (no connect).
C1	UEX_RST	Digital Output	Semtech UEX USB 2.0 extender Reset. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
C2	UEX_BOOT_SELO	Digital Output	Semtech UEX USB 2.0 extender UART program enable. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
C3	UEX_UART_TX	Digital Output	Semtech UEX USB 2.0 extender UART. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
C4	UEX_PHY2LEN	Digital Output	Semtech UEX USB 2.0 extender Layer 2 enable. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
C7	UEX_TX_CLK	Digital Output	Semtech UEX USB 2.0 extender transmit clock. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
C8	UEX_MDIO	Digital Bi-Directional	Semtech UEX USB 2.0 extender MDIO. This ball has an internal weak pull-up resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
C9	HDMI_RX_HPD	Digital Output	HDMI Rx HPD (Hot Plug Detect). Inform the HDMI source that a sink is present.
C10	HDMI_RX_REXT	Analog Bi-Directional	HDMI Rx PHY 1.62kΩ precision resistor to ground.
C14, C13	HDMI_TX_REC_CLK_P, HDMI_TX_REC_CLK_N	Input	HDMI Tx recovered clock.
C17	HDMI_TX_CEC	Digital Open Drain	HDMI Tx CEC (Consumer Electronics Control). Used to transmit control over HDMI cable. When CEC messages are sent using the API, CEC logical address 14 (0x0E - free use) is used by default. Note: CEC commands can be sent over the IP network using the SDVoE API. However, when two devices are connected in Point-to-Point mode (no Ethernet switch), CEC is automatically paired across the 10G connection. In this mode, the device will not respond to CEC API commands.
C21	HDMI_TX_5V	Digital Output	HDMI Tx +5V signal. Note that this is a 3.3V output and a level shifter is required to drive 5V logic. Inform the sink module that a source is available. This output is permanently set high. Note: The state of this output can be overridden using the SDVoE API.
C22	HDMI_TX_HPD	Digital Input	HDMI Tx HPD (Hot Plug Detect). Inform the HDMI source that a sink is present. 5V tolerant.
D2, F5, E2, E5, E3, F4, F2, E1, G5, H5, D1, E4	GPIO_[11:0]	Digital Bi-Directional	General purpose IO. These balls have internal weak pull-down resistors.
D4	UEX_UART_RX	Digital Input	Semtech UEX USB 2.0 extender UART debug. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
D5	UEX_HOST	Digital Input	Semtech UEX USB 2.0 extender STATUS. This ball has an internal weak pull-up resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
D6	UEX_RX_CLK	Digital Input	Semtech UEX USB 2.0 extender receive clock. When the UEX USB 2.0 extender interface is not used, this ball should be connected to ground.
D7, D8, E8, C6	UEX_RX_DATA [3:0]	Digital Input	Semtech UEX USB 2.0 extender RGMII Receive Data. When the UEX USB 2.0 extender interface is not used, this ball should be connected to ground.
D15	HDMI_RX_5V	Digital Input	HDMI Rx +5V signal. Inform the HDMI Rx module that a source is available. 5V tolerant.

Table 1-1: Ball Descriptions

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Ball Number	Name	Type	Description
D16	HDMI_RX_CEC	Digital Open Drain	HDMI Rx CEC (Consumer Electronics Control). Used to transmit control over HDMI cable. Note: CEC commands can be sent over the IP network using the SDVoE API. However, when two devices are connected in Point-to-Point mode (no Ethernet switch), CEC is automatically paired across the 10G connection. In this mode, the device will not respond to CEC API commands.
D20	HDMI_TX_REXT	Analog Bi-Directional	HDMI Tx PHY 1.62kΩ precision resistor to ground.
E6	UEX_TX_CTL	Digital Output	Semtech UEX USB 2.0 extender transmit data enable. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
E7	UEX_RX_CTL	Digital Input	Semtech UEX USB 2.0 extender receive data enable. When the UEX USB 2.0 extender interface is not used, this ball should be connected to ground.
E9	UEX_MDC	Digital Input	Semtech UEX USB 2.0 extender MDIO. This ball has an internal weak pull-up resistor. When the UEX USB 2.0 extender interface is not used, this ball should be left open (no connect).
E13	HDMI_RX_DDC_SCL	Digital Open Drain	HDMI Rx DDC SCL. 5V tolerant. When the HDMI RX interface is not used, this ball should be connected to ground.
E14	HDMI_RX_DDC_SDA	Digital Open Drain	HDMI Rx DDC SDA. 5V tolerant. When the HDMI RX interface is not used, this ball should be connected to ground.
E16	HDMI_TX_DDC_SCL	Digital Open Drain	HDMI Tx DDC SCL. 5V tolerant. When the HDMI TX interface is not used, this ball should be connected to ground.
E17	HDMI_TX_DDC_SDA	Digital Open Drain	HDMI Tx DDC SDA. 5V tolerant. When the HDMI TX interface is not used, this ball should be connected to ground.
F1, H1, G2, T3, T4, U5	RSVD_6, RSVD_5, RSVD_4, RSVD_2, RSVD_1, RSVD_0	NC	Reserved. Do not connect.
G1	CLK_MNG_FINC_FB	Digital Input	Clock Management Frequency Increment Feedback input. This ball is only used for designs using the Renesas RC22504A200GNK clock chip. Connect directly to CLK_MNG_FINC (ball AB16). For all other clock devices, this input is reserved and should be left unconnected.
G3	SINGLE_I2S_SPI_SCLK	Digital Output	Single I ² S ADC/DAC control SPI serial clock. Only Cirrus Logic WM8731SEDS supported. When an alternate audio CODEC device is used, it must be controlled by an external MCU. See PDS-062730. When Single I ² S interface is unused, this ball should be left open.

Table 1-1: Ball Descriptions

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Ball Number	Name	Type	Description
G4	SINGLE_I2S_SPI_CSB	Digital Output	Single I ² S ADC/DAC control SPI chip select. Only Cirrus Logic WM8731SEDS supported. When an alternate audio CODEC device is used, it must be controlled by an external MCU. See PDS-062730. When Single I ² S interface is unused, this ball should be left open.
G23, J19, H20, G21, K20, F22, H21, J22, F21, G19, C24, E23, D23, H19, E22, G20, V23, T19, V20, U22, V21, W23, W22, U21, V19, W24, Y23, AB24, Y22, AA21, W20, AB23	DDR3_DQ_[31:0]	Digital Bi-Directional	DDR3 Data input/output. Bidirectional data bus for the x4 configuration. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
H3	CLK_MNG_FDEC_FB	Digital Input	Clock Management Frequency Decrement Feedback input. This ball is only used for designs using the Renesas RC22504A200GNK clock chip. Connect directly to CLK_MNG_FDEC (ball AC16). For all other clock devices, this input is reserved and should be left unconnected.
H4	SINGLE_I2S_SPI_SDIN	Digital Output	Single I ² S ADC/DAC control SPI serial data input. Only Cirrus Logic WM8731SEDS supported. When an alternate audio CODEC device is used, it must be controlled by an external MCU. See PDS-062730. When Single I ² S interface is unused, this ball should be left open.
H22, AB22, L19	DDR3_VREF_[2:0]	Analog Bi-Directional	Vref for Receiver of DDR PHY. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
H24, H23 F23, F24 U24, U23 AA23, AA24	DDR3_DQS_3_P, DDR3_DQS_3_N DDR3_DQS_2_P, DDR3_DQS_2_N DDR3_DQS_1_P, DDR3_DQS_1_N DDR3_DQS_0_P, DDR3_DQS_0_N	Digital Bi-Directional	DDR3 Data strobe. Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
J5	LED_USB_HID_ACTIVITY	Digital Output	USB HID Activity LED off: Either USB HID is not enabled or not presently paired, or paired with a device in OTG mode with no USB device connected (no active role selected). LED blinking: USB HID is paired but no local USB host device or remote USB device is connected. LED on: indicates USD HID is paired plus both a local USB host and a remote USB device are connected. LED flashes twice followed by off for 1 second: when two paired devices have encryption enabled, indicates that authentication failed.

Table 1-1: Ball Descriptions

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Ball Number	Name	Type	Description
J2	LED_VIDEO_OUT_STABLE	Digital Output	Video Out Stable Reflects the status of the Source Stable signal received from a Transmitter as transported over the 10GbE link to the Receiver. This LED is also impacted by the current video mode. When a Rx is operating in: Genlock, Genlock Scaling or Genlock Wall Mode, it follows the status of the Source Stable received from the transmitter. Fast Switch, Multiview or Fast Switch Wall Mode, it is always ON. This is because Rx is always driving a valid HDMI output.
K4	LED_USB_ACTIVITY	Digital Output	Semtech UEX USB 2.0 extender USB Activity LED blinking Indicates USB data traffic present.
J1	LED_USB_PRESENT	Digital Output	Semtech UEX USB 2.0 extender USB Present LED illuminated if UEX extender is detected.
K3	LED_10G_RX_ACTIVITY	Digital Output	10G Rx LED Indicates if Ethernet packets are received. Also, indicates if XFI interface detects valid signal. LED off: Packets are not received and the Rx SERDES is NOT linked. LED blinking: Packets are received and the Rx SERDES is linked. LED on: Packets are not transmitted and the Rx SERDES is linked.
J3	LED_10G_TX_ACTIVITY	Digital Output	10G Tx LED Indicates if Ethernet packets are transmitted. LED off: Packets are not transmitted LED blinking: Packets are transmitted.
K1	LED_SYSTEM_STATUS	Digital Output	System Status LED Steady on: Main application firmware is running. Slow blinking: Golden firmware is running. Fast blinking: Bootloader code is running.
K5	LED_VIDEO_IN_STABLE	Digital Output	Video In Stable LED LED illuminated if video signal running through the BlueRiver chipset is stable. Stability is established when at least two consecutive video frames have the same horizontal and vertical active dimensions.
J11	AVDD18_HDMIRX	Power	HDMI Rx analog supply voltage (1.8V).
J12	AVDD33_HDMIRX	Power	HDMI Rx analog supply voltage (3.3V).
J13, M11, N11, P13	VDD33	Power	3.3V supply for 5V tolerant open drain (CEC, DDC, HPD)
J14	AVDD18_LVDS	Power	1.8V supply for LVDS IO.
J15	AVDD18_HDMITX	Power	HDMI Tx analog supply voltage (1.8V).
K11	AVDD09_HDMIRX	Power	HDMI Rx analog supply voltage (0.9V).
K13	AVDD09_HDMITX	Power	HDMI Tx analog supply voltage (0.9V).
K15, P15	VDD18_DDR	Power	DDR interface pre-driver (1.8V) power supply.

Table 1-1: Ball Descriptions

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Ball Number	Name	Type	Description
K19, D24, W21, Y21	DDR3_DM_[3:0]	Output	DDR3 Input data mask. DM is a mask signal for write data. Input data is masked when DM is sampled HIGH along with the input data during a write access. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
K21	DDR3_RESET_N	Output	DDR3 Reset. Active low. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
L2	IR_DATA_IN	Digital Input	Infrared input. This ball has an internal weak pull-down resistor. When Infrared interface is unused, this ball should be left open.
L3	IR_DATA_OUT	Digital Output	Infrared output. When Infrared interface is unused, this ball should be left open.
L4, M1, M5, L1	QUAD_I2S_SD_OUT [3:0]	Digital Output	QUAD I2S data out. When Quad I2S interface is unused, this ball should be left open.
L5	QUAD_I2S_WS_OUT	Digital Output	QUAD I2S word select out. When Quad I2S interface is unused, this ball should be left open.
L11	VDD_UEX	Power	Semtech UEX USB 2.0 extender RGMII IO power supply (1.8V). Whether the UEX USB 2.0 extender interface is used or not, this supply shall be connected to 1.8V.
L12, N12	VDD09	Power	Core logic power supply (0.9V).
L14, M13, P14	VDD09_DDR	Power	DDR Core power supply (0.9V).
L15, M15, N15	VDD_DDR	Power	DDR IO power supply (1.5V or 1.35V). Connect to 0.9V when DDR is not populated. Connect to 1.35V if a 1.35V RAM is used.
L22, M21, R23	DDR3_BA [2:0]	Digital Output	Bank address inputs. BA[2:0] define the bank to which an ACTIVATE, READ, WRITE, or PRECHARGE command is being applied. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
L23, M20	DDR3_ZQ_05_P, DDR3_ZQ_05_N	Digital Bi-Directional	Reference resistor for Calibration. Connect positive lane (L23) to ground via a 240Ω pull-down resistor. The negative lane (M20) should be left open (no connect). (AVP2000/AVP2000T) These balls are Reserved in AVP1000 where M20 should be left open and L23 connected to ground via a 240Ω pull-down resistor.
M2	QUAD_I2S_SCK_IN	Digital Input	QUAD I2S serial clock in. This ball has an internal weak pull-up resistor. When Quad I2S interface is unused, this ball should be left open.

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Ball Number	Name	Type	Description
M24	DDR3_CKE	Digital Output	DDR3 Clock enable. CKE enables (registered HIGH) and disables (registered LOW) internal circuitry and clocks on the DRAM. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
N2	QUAD_I2S_SCK_OUT	Digital Output	QUAD I2S serial clock out. When Quad I2S interface is unused, this ball should be left open.
N3	QUAD_I2S_MCLK_OUT	Digital Output	QUAD I2S master clock out. When Quad I2S interface is unused, this ball should be left open.
N4	QUAD_I2S_WS_IN	Digital Input	QUAD I2S word select in. This ball has an internal weak pull-up resistor. When Quad I2S interface is unused, this ball should be left open.
N5	QUAD_I2S_MCLK_IN	Digital Input	QUAD I2S master clock in. This ball has an internal weak pull-up resistor. When Quad I2S interface is unused, this ball should be left open.
N14	AVDD18_DDR_PLL	Power	DDR PLL analog supply voltage (1.8V)
N20	DDR3_CS_N	Digital Output	DDR3 Chip select. CS_N enables (registered LOW) and disables (registered HIGH) the command decoder. All commands are masked when CS_N is registered HIGH. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
N21	DDR3_ODT	Digital Output	DDR3 On-die termination. ODT enables (registered HIGH) and disables (registered LOW) termination resistance internal to the DDR3 SDRAM. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
N22	DDR3_CAS_N	Digital Output	DDR3 Command. Column Access Strobe (CAS). Active low. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
N23, N24	DDR3_CK_P, DDR3_CK_N	Digital Output	Differential clock outputs. All address and control input signals are sampled on the crossing of the positive edge of CK_P and the negative edge of CK_N. Input data strobe is referenced to the crossings of CK_P and CK_N. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
P1	REG_SDA	Digital Open Drain	Register I2C access (SDA). This ball has an internal weak pull-up resistor. When the I2C interface is not used, this ball should be left open (no connect).

Table 1-1: Ball Descriptions

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Ball Number	Name	Type	Description
P4, M4, N1, P3	QUAD_I2S_SD_IN [3:0]	Digital Input	QUAD I2S data in. These balls have internal weak pull-up resistors. When Quad I2S interface is unused, this ball should be left open.
P5	SFP_SDA	Digital Open Drain	SFP control I2C interface SDA. This ball has an internal weak pull-up resistor.
P10	AVDD18_PLL	Power	PLL analog supply voltage (1.8V).
P11	AVDD09_CML	Power	XFI PHY reference clock power supply.
P20	DDR3_RAS_N	Digital Output	DDR3 Command. Row Access Strobe (RAS). Active low. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
P22	DDR3_WE_N	Digital Output	DDR3 Command. Write Enable (WE). Active low. (AVP2000/AVP2000T) This ball is Reserved in AVP1000 and should be left open (no connect).
R1	REG_SCL	Digital Open Drain	Register I2C access (SCL). This ball has an internal weak pull-up resistor. When the I2C interface is not used, this ball should be left open (no connect).
R2	SFP_TX_DISABLE	Digital Output	SFP Tx disable.
R3	SFP_RATE_SEL	Digital Output	SFP rate select.
R5	SFP_SCL	Digital Open Drain	SFP control I2C interface SCL. This ball has an internal weak pull-up resistor.
R10	AVDD09_XFI	Power	XFI analog supply voltage (0.9V).
R11	AVDD18_XFI	Power	XFI analog supply voltage (1.8V).
R12	VDD_PHY1G	Power	1G RGMII IO power supply. Can be driven by a 2.5V or a 1.8V supply. When the 1G RGMII interface is not used, this supply can be connected to 1.8V to reduce power.
R14	VDD18	Power	IO power supply (1.8V).
T1	SFP_LOSS	Digital Input	SFP Loss.
T2	SFP_PRESENT_N	Digital Input	SFP Rate present. Active low. This ball has an internal pull-up resistor.
T5	SINGLE_I2S_DIR	Digital Output	Single I2S direction. When Single I2S interface is unused, this ball should be left open.
T12	VDD_PHY10G	Power	10G MDIO IO power supply. Can be driven by a 3.3V, a 2.5V or a 1.8V supply.
T15	VDD18_JTM	Power	Temperature sensor analog supply (1.8V).

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Ball Number	Name	Type	Description
T20, T22	DDR3_ZQ_08_P, DDR3_ZQ_08_N	Analog Bi- Directional, Analog Input	Reference resistor for Calibration. Connect positive lane (T20) to ground via a 240Ω pull-down resistor. The negative lane (T22) should be left open (no connect). These balls are Reserved in AVP1000 where T22 should be left open and T20 connected to ground via a 240Ω pull-down resistor.
T23, R21, L21, J24, M23, R20, M19, U20, J21, R22, K24, R24, K22, K23, P19	DDR3_ADDR [14:0]	Digital Output	Address. Provide the row address for ACTIVATE commands, and the column address and auto pre-charge. (AVP2000/AVP2000T) These balls are Reserved in AVP1000 and should be left open (no connect).
U1	SINGLE_I2S_LRC	Digital Output	Single I2S left right clock. When Single I2S interface is unused, this ball should be left open.
U2	SINGLE_I2S_BCLK	Digital Output	Single I2S bit clock. When Single I2S interface is unused, this ball should be left open.
U4	SINGLE_I2S_SD_OUT	Digital Output	Single I2S data out. When Single I2S interface is unused, this ball should be left open.
V1	SINGLE_I2S_SD_IN	Digital Input	Single I2S data in. This ball has an internal weak pull-up resistor. When Single I2S interface is unused, this ball should be left open.
V2	SINGLE_I2S_MCLK	Digital Output	Single I2S master clock. When Single I2S interface is unused, this ball should be left open.
V5, V3, V4, W4, Y7, AC5, AA8, AB6	ULPI_DATA [7:0]	Digital Bi- Directional	ULPI data to/from link. When the USB-HID interface is not used, this ball should be left open (no connect).
W1	ULPI_STP	Digital Output	ULPI stop signal. When the USB-HID interface is not used, this ball should be left open (no connect).
W2	ULPI_CLOCK	Digital Input	ULPI clock. When the USB-HID interface is not used, this ball should be connected to ground.
W3	ULPI_NXT	Digital Input	ULPI next signal. When the USB-HID interface is not used, this ball should be connected to ground.
Y3	ULPI_DIR	Digital Input	ULPI direction. When the USB-HID interface is not used, this ball should be connected to ground.
Y4	ULPI_RESET_N	Digital Output	ULPI PHY reset. Active LOW reset. When the USB-HID interface is not used, this ball should be left open (no connect).

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
Y6	XFI_TX0_POLARITY	Digital Input	XFI Tx polarity configuration. '0' = normal polarity '1' = inverted polarity This ball has an internal weak pull-down resistor.
Y8	PHY_10G_INT_N	Digital Input (2.5V or 3.3V)	Interrupt signal from 10G PHY. Active low. This ball has an internal weak pull-up resistor.
Y12	UART_2_RX	Digital Input	UART #2 In. This ball has an internal weak pull-up resistor.
Y13	UART_2_TX	Digital Output	UART #2 Out.
Y14	FW_FLASH_SCK	Digital Output	SPI FLASH serial clock. Connect to external SPI FLASH used for device firmware.
Y15	CLK_MNG_SCL	Digital Open Drain	Clock Management I2C interface clock. This ball has an internal weak pull-up resistor. When the device is configured as a Transmitter, this ball is unused and should be left open (no connect).
Y16	CLK_MNG_SDA	Digital Open Drain	Clock Management I2C interface data. This ball has an internal weak pull-up resistor. When the device is configured as a Transmitter, this ball is unused and should be left open (no connect).
Y17	IREF	Analog	Reserved. Connect 10kΩ pull-down resistor to GND.
AA1, AA2	RSVD [20:19]	Analog Input	Reserved. Do not connect.
AA6	XFI_RX0_POLARITY	Digital Input	XFI Rx polarity configuration. '0' = normal polarity '1' = inverted polarity This ball has an internal weak pull-down resistor.
AA7	PHY_10G_MDC	Digital Output (2.5V or 3.3V)	MDIO clock.
AA9, Y9, AB8, AC8	PHY_1G_TX_DATA [3:0]	Digital Output (1.8V or 2.5V)	1G PHY RGMII transmit data bus. Must be connected to the RGMII RX balls of the external switch. When 1G interface is unused, this ball should be left open.
AA11	PHY_1G_RX_CLK	Digital Input (1.8V or 2.5V)	PHY 1G RGMII Tx clock. Operates at 125MHz. When 1G interface is unused, this ball should be left open.
AA12	UART_4_RX	Digital Input	UART #4 In. This ball has an internal weak pull-up resistor.
AA14	FW_FLASH_IO3	Digital Bi-Directional	IO3 when in Quad mode, unused when SPI. SPI FLASH Quad Instructions Data Input Output 3. Connect to external SPI FLASH used for device firmware.
AA15	INT	Digital Output	Interrupt to signal an event to an external component.
AA17, AC18	A_SEL [1:0]	Digital Input	Used to select 2 LSB of I ² C register slave address. This ball has an internal weak pull-up resistor.

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
AA18, AB19, AC19, AD17, AD18, AB18, Y18, AD19	RSVD [15:8]	NC	Reserved for Semtech use. Do not connect.
H16	RSVD [16]	Digital Input	Reserved for Semtech use. Recommend connecting to GND.
AC22, AC23	RSVD [18:17]	NC	Reserved for Semtech use. Do not connect.
AB4, AB3	XFI_RX0_P, XFI_RX0_N	Analog/PHY Input	XFI Serial In.
AB10	PHY_1G_RX_CTL	Digital Input (1.8V or 2.5V)	PHY 1G RGMII Tx CTL. When 1G interface is unused, this ball should be left open.
AB11	UART_3_RX	Digital Input	UART #3 In. This ball has an internal weak pull-up resistor.
AB12	UART_4_TX	Digital Output	UART #4 Out.
AB13	FW_FLASH_CS_N	Digital Output	SPI Flash active low chip select. Connect to external SPI FLASH used for device firmware.
AB15	CLK_MNG_RESET_N	Digital Output	Clock Management IC reset. Active low.
AB16	CLK_MNG_FINC	Digital Output	Clock Management frequency increment.
AB17	RST_N	Digital Input	Main Reset. Active low.
AB20	TCK	Digital Input (1.8V)	Test clock input ball of the IEEE-1149.1 Test access port (TAP).
AB21	TRST	Digital Input (1.8V)	Test reset input ball of the IEEE-1149.1 Test access port (TAP). The reset is active low.
AC2, AC1	XFI_TX0_P, XFI_TX0_N	Analog/PHY Output	XFI Serial Out.
AC6	PHY_10G_RST_OUT_N	Digital Output (2.5V or 3.3V)	External 10GBaseT PHY reset. Active low.
AC7	PHY_10G_TX_EN	Digital Output (2.5V or 3.3V)	When pulled low, this ball disables the output line drivers on the 10GBaseT PHY.
AC9	PHY_1G_TX_CLK	Digital Output (1.8V or 2.5V)	PHY 1G RGMII Rx CLK. Operates at 125 MHz. When 1G interface is unused, this ball should be left open.
AC11	UART_3_TX	Digital Output	UART #3 Out.
AC12	UART_1_RX	Digital Input	UART #1 In. This ball has an internal weak pull-up resistor. Note: when using the Semtech UEX USB 2.0 extender with the device, if UEX first time programming is required using the ProAV Programming Tool, UART_1 must be used. See the Semtech_UEX_Attached_to_BlueRiver_AVP_Manufacturer_Guide (PDS-063010) for more details.

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
AC13	BOOT_MODE	Digital Input	Boot mode: '0' = Normal operation '1' = Internal ROM This ball has an internal weak pull-down resistor.
AC14	FW_FLASH_SI_IO0	Digital Bi-Directional	Serial Input for single bit data commands or IO0 for Dual or Quad commands.
AC15	FACTORY	Digital Input	Set all settings to default value when FACTORY is set from 0 to 1. FACTORY should stay 1 for at least 100ms.
AC16	CLK_MNG_FDEC	Digital Output	Clock Management frequency decrement.
AC17	ULPI_VBUS	Digital Input	ULPI VBUS state. Should be connected to VBUS ball of the USB HID ULPI PHY. Connect to GND if not used.
AC20	TDO	Digital Output (1.8V)	Test data output ball of the IEEE-1149.1 Test access port (TAP).
AC21	TDI	Digital Input (1.8V)	Test data input ball of the IEEE-1149.1 Test access port (TAP).
AD4	XFI_REXT	Analog/PHY Bi-Directional	Reserved. Do not connect.
AD5	ULPI_CS_N	Digital Output	When CS_N is de-asserted, all balls at ULPI interface are in tristate. Active low. When the USB-HID interface is not used, this ball should be left open (no connect).
AD7	PHY_10G_MDIO	Digital Bi-Directional (2.5V or 3.3V)	MDIO data line. This ball has an internal weak pull-up resistor.
AD9	PHY_1G_TX_CTL	Digital Output (1.8V or 2.5V)	PHY 1G RGMII Rx CTL. When 1G interface is unused, this ball should be left open.
AD11, Y11, Y10, AC10	PHY_1G_RX_DATA [3:0]	Digital Input (1.8V or 2.5V)	1G PHY RGMII receive data bus. Must be connected to the RGMII TX balls of the external switch. These balls have internal weak pull-up resistors. When 1G interface is unused, these balls should be left open.
AD12	UART_1_TX	Digital Output	UART #1 Out. Note: when using the Semtech UEX USB 2.0 extender with the device, if first time programming is required using the ProAV Programming Tool, UART_1 must be used. See the Semtech_UEX_Attached_to_BlueRiver_AVP_Manufacturer_Guide (PDS-063010) for more details.
AD14	FW_FLASH_SO_IO1	Digital Bi-Directional	SPI FLASH Standard Instructions Data Output (Dual and Quad Instructions Data Input Output 1). Connect to external SPI FLASH used for device firmware.
AD15	FW_FLASH_IO2	Digital Bi-Directional	SPI FLASH Quad Instructions Data Input Output 2. Connect to external SPI FLASH used for device firmware.
AD21	XTAL_25_IN	Digital Input (1.8V)	25MHz crystal input reference clock.

Table 1-1: Ball Descriptions

All Digital I/O operate at 3.3V unless otherwise noted in the Type column. All DDR-related Digital I/O are selectable to operate at 1.35V or 1.5V.

Ball Number	Name	Type	Description
AD22	XTAL_25_OUT	Digital Output (1.8V)	25MHz crystal output reference clock.
AD23	TMS	Digital Input (1.8V)	Test mode select input ball of the IEEE-1149.1 Test access port (TAP).

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ADV
02/03/2026

2 Electrical Characteristics

2.1 Absolute Maximum Ratings

Table 2-1: Absolute Maximum Ratings

Parameter	Value	Units
Digital I/O Supply Voltage 3.3	-0.3 to 4.0	V
Digital I/O supply voltage 2.5	-0.3 to 3.5	V
Digital I/O supply voltage 1.8	-0.3 to 1.98	V
Digital Core supply voltage	-0.3 to 1.1	V
Analog Core 2.5V supply voltage	-0.3 to 3.5	V
Analog Core 1.8V supply voltage	-0.3 to 1.98	V
Analog Core 0.9V supply	-0.3 to 1.1	V
SSTL 13 supply voltage	-0.3 to 1.8	V
Temperature operating range (junction)	$-20 < T_J \leq 100$	°C
Storage temperature range	$-50 < T_S < 125$	°C
ESD voltage (any ball) HBM	1000	V
Lead temperature (soldering: 4 sec.)	260	°C

2.2 DC Electrical Characteristics

Table 2-2: Power

Parameter	Symbol	Conditions	Min	Typ1	Max2	Units	Notes
Supply Current							
VDD09 & AVDD09 Supply Current	IVDD09	0.9V	—	1905	2798	mA	—
VDD_DDR Supply Current	IVDD_DDR	0.9V no DDR	—	37	44	mA	3
VDD_DDR Supply Current	IVDD_DDR	1.35V with DDR	—	116	141	mA	4
VDD_DDR Supply Current	IVDD_DDR	1.5V with DDR	—	129	157	mA	5
VDD18 & AVDD18 Supply Current	IVDD18	1.8V	—	156	165	mA	—
VDD_PHY10G Supply Current	IVDD_PHY10G	1.8V	—	3	6	mA	—
VDD_PHY10G Supply Current	IVDD_PHY10G	2.5V	—	4	8	mA	—
VDD_PHY10G Supply Current	IVDD_PHY10G	3.3V	—	6	10	mA	—
VDD_PHY1G Supply Current	IVDD_PHY1G	1.8V	—	6	22	mA	—
VDD_PHY1G Supply Current	IVDD_PHY1G	2.5V	—	9	31	mA	—
VDD_UEX Supply Current	IVDD_UEX	1.8V	—	6	22	mA	—
VDD33 & AVDD33 Supply Current	IVDD33	3.3V	—	141	149	mA	6
Total Power	—	—	—	2.65	—	W	7

Notes:

1. Typical supply currents measured at TA = 22°C with HDMI input and output at 4096x2160p 60Hz with random dot pattern. 1G interface using 100Mb/s unless otherwise noted.
2. Maximum supply numbers measured at maximum temperature, Maximum voltages with 1G and UEX interfaces at 940 Mb/s.
3. No DDR3 installed with VDD_DDR connected to 0.9V.
4. DDR3 installed with VDD_DDR connected to 1.35V supply.
5. DDR3 installed with VDD_DDR connected to 1.5V supply.
6. Excluding VDD_UEX & VDD_PHY10G balls.
7. DDR3 installed. VDD_DDR balls connected to 1.35V and VDD_PHY10G balls connected to 2.5V.
8. All 0.9V rails should be within ±5% voltage tolerance range.

Table 2-3: DC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
HDMI Rx (TMDS Sink)							
Supply Voltage	AVDD33_HDMIRX	—	3.135	3.3	3.465	V	—
Supply Voltage	AVDD18_HDMIRX	—	1.710	1.8	1.890	V	—
Supply Voltage	AVDD09_HDMIRX	—	0.855	0.9	0.945	V	—
Sink Termination Impedance		—	90	110	121	Ω	—
Input Differential Voltage	Vidiff	—	150	—	1200	mV	—
Input Common Mode Voltage	Vicm1	—	AVDD33_HDMIRX - 700	—	AVDD33_HDMIRX - 37.5	mV	—
	Vicm2	—	AVDD33_HDMIRX - 10	—	AVDD33_HDMIRX + 10	mV	—
Through Connection Impedance		—	85	100	115	Ω	—
HDMI Tx (TMDS Source)							
Supply Voltage	AVDD18_HDMITX	—	1.710	1.8	1.890	V	—
Supply Voltage	AVDD09_HDMITX	—	0.855	0.9	0.945	V	—
Source Termination Impedance		—	75	—	150	Ω	—
Through Connection Impedance		—	85	100	115	Ω	—
Single-Ended High Level Voltage Range: Data Channels 0, 1, 2		—	AVCC - 400	—	AVCC + 10	mV	1
Single-Ended Low Level Voltage Range: Data Channels 0, 1, 2		—	AVCC - 1000	—	AVCC - 400	mV	1
Single-Ended High Level Voltage Range: Clock Channel		—	AVCC - 400	—	AVCC + 10	mV	1
Single-Ended Low Level Voltage Range: Clock Channel		—	AVCC - 1000	—	AVCC - 200	mV	1
Single-Ended Swing Voltage: Data Channels 0, 1, 2		—	400	—	600	mV	—
Single-Ended Swing Voltage: Clock Channel		—	200	—	600	mV	—
DDR3 (1.35V)							

Table 2-3: DC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Supply Voltage		—	1.283	1.35	1.45	V	—
I/O Supply Voltage		—	1.283	1.35	1.45	V	—
Input Leakage Current		—	-2	—	2	μA	—
Input Low Voltage		—	VSS	—	—	V	—
Input High Voltage		—	—	—	VDD_DDR	V	—
DDR3 (1.5V)							
Supply Voltage		—	1.425	1.5	1.575	V	—
I/O Supply Voltage		—	1.425	1.5	1.575	V	—
Input Leakage Current		—	-2	—	2	μA	—
Input Low Voltage		—	VSS	—	—	V	—
Input High Voltage		—	—	—	VDD_DDR	V	—
Digital IO							
Supply Voltage (3.3V Operation)		—	3	3.3	3.6	V	—
Input High Voltage (3.3V Operation)	VIH	—	2	—	3.3	V	—
Input Low Voltage (3.3V Operation)	VIL	—	-0.5	—	0.8	V	—
Output High Voltage (3.3V Operation)	VOH	IOH = 12mA	2.4	—	—	V	—
Output Low Voltage (3.3V Operation)	VOL	IOL = 12mA	—	—	0.4	V	—
Supply Voltage (1.8V Operation)		—	1.62	1.8	1.98	V	—
Input High Voltage (1.8V Operation)	VIH	—	1.4	—	1.98	V	—
Input Low Voltage (1.8V Operation)	VIL	—	-0.3	—	0.4	V	—
Output Voltage (1.8V Operation)	VO	IO = 100μA	0	—	1.8	V	—
I2C							
LOW level input voltage	VIL	Fast Mode	-0.5	—	0.3*VDD33	V	—
HIGH level input voltage	VIH	—	0.7*VDD33	—	—	V	—
LOW level output voltage		—	0	—	0.4	V	—

Table 2-3: DC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
LOW level output current		—	3	—	—	mA	—
MDIO (10Gb/s)							
Input High Voltage		—	0.84	1.2	1.5	V	—
Input Low Voltage		—	-0.3	—	0.36	V	—
Output High Voltage		IOH = -100µA	1	—	1.5	V	—
Output High Voltage		IOL = 100µA	-0.3	—	0.2	V	—
Output High Current		Vi = 1V	—	—	-4	mA	—
Output Low Current		Vi = 2V	4	—	—	mA	—
XFI							
Tx DC Common Mode Voltage		—	0	—	3.6	V	—
ESD HBM		—	1K	—	—	V	—
ESD CDM		—	250	—	—	V	—
Latch Up		—	+/- 100	—	—	mA	—
RGMII (2.5V) [2]							
Core Supply Voltage	VDD_PHY1G	—	2.25	2.5	2.75	V	—
IO Supply Voltage	VDD_PHY1G	—	—	2.5	—	V	—
Voltage at IO		—	0	—	VDD_PHY1G	V	—
Input Logic High	VIH	2.5V IO supply	1.7	—	—	V	—
Input Logic Low	VIL	2.5V IO supply	—	—	0.7	V	—
Output Logic High @ IOH = 100µA	VOH	2.5V IO supply	2.0	—	VDD_PHY1G + 0.3	V	—
Output Logic Low @ IOL = 100µA	VOL	2.5V IO supply	VSS - 0.3	—	0.4	V	—
RGMII (1.8V) [2, 3]							
Core Supply Voltage	VDD_PHY1G VDD_UEX	—	1.8	—	1.89	V	4
IO Supply Voltage	VDD_PHY1G VDD_UEX	—	1.71	1.8	1.89	V	5
Voltage at IO		—	0	—	VDD_PHY1G	V	—
Input Logic High	VIH	1.8V IO supply	0.7* VDD_PHY1G	—	VDD_PHY1G+0.3	V	—

Table 2-3: DC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Input Logic Low	VIL	1.8V IO supply	VSS-0.3	—	0.3* VDD_PHY1G	V	—
Output Logic High @ IoH = 100μA	VOH	1.8V IO supply	0.85* VDD_PHY1G	—	VDD_PHY1G + 0.3	V	—
Output Logic Low @ IoL = 100μA	VOL	1.8V IO supply	VSS - 0.3	—	0.15* VDD_PHY1G	V	—

Notes:

1. AVCC represents the termination supply on the receiving side.
2. 2.5V does not apply for Semtech UEX USB 2.0 extender RGMII. The UEX RGMII can only operate at 1.8V. The 1G RGMII can only operate at 2.5V or 1.8V.
3. 1G RGMII operation at 1.8V requires output ball with name starting with PHY_1G_* to have a maximum load of 10pF.
4. When the PHY_1G interface is enabled, VDD_PHY1G should not be lower than 1.8V
5. When the PHY_1G interface is not used, power supply with typical value of 1.8V±5% can be used.

2.2.1 Power Supply Noise

Noise on the DC power supplies may affect the performance of high-frequency circuits. It is imperative to use proven noise filtering techniques to meet noise requirements. The use of low DCR ferrite beads is recommended. The power balls starting with an “A” are analog supply balls and care must be taken to meet or be lower than the maximum power supply noise levels shown in the following table.

Table 2-4: Power Supply Maximum Noise Levels

Supply Name	DC Supply Voltage (V)	Maximum Noise (mV _{pp})
VDD18/JTM	1.8	30
AVDD18_XFI	1.8	30
VDD09	0.9	35
VDD_DDR	1.35/1.5	50
AVDD09_XFI/XML	0.9	25
AVDD09_DDR	0.9	25
AVDD09_HDMIRX	0.9	25
AVDD09_HDMITX	0.9	25
VDD18_DDR	1.8	25
AVDD18_DDR_PLL	1.8	25
AVDD18_PLL	1.8	25
AVDD18_HDMIRX	1.8	25
AVDD18_HDMITX	1.8	25
AVDD18_LVDS	1.8	25
AVDD33_HDMIRX	3.3	30
VDD33	3.3	50

For the recommended power supply noise decoupling components, please refer to PDS-062051 *Hardware Design Guide*.

2.2.2 Power Supply Sequencing

The following power supply rail sequencing requirements must be met:

- The 0.9V core supply must reach maximum rail before (timing not critical) or at the same time as the 3.3V digital I/O supply reaches maximum rail.
- Required voltage rail slew rate: $200\mu\text{s} < T_{\text{SLEW}} < 50\text{ms}$ for all power supplies.
- Time from the first power supply ramp starting point to the last supply ramp end point must be within T_{SLEW} specification of $< 50\text{ms}$.
- Ramp up of each power supply must not have a plateau or dip (must increase monotonically).

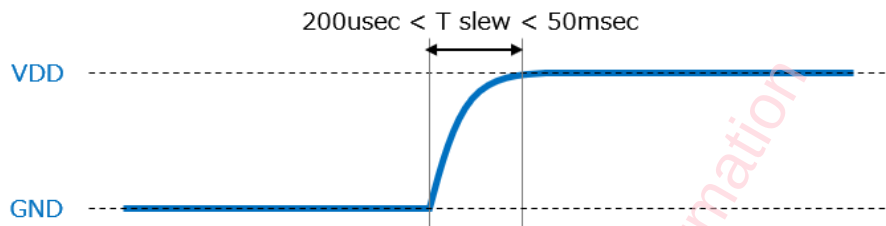


Figure 2-1: Power Supply Ramp Specification

2.3 AC Electrical Characteristics

Table 2-5: AC Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
HDMI Rx (TMDS Sink)							
Data Rate		—	0.25	—	6	Gb/s	—
Minimum differential sensitivity		—	150	—	—	mV	—
Max Allowable Intra-pair skew		—	—	—	$0.15T_{\text{bit}} + 112\text{ps}$	—	—
Max allowable Inter-pair skew		—	—	—	$0.2T_{\text{character}} + 1.78\text{ns}$	—	—
TMDS Clock Jitter		Relative to Ideal Recovery Clock	—	—	$0.3 T_{\text{bit}}$	—	—
TDR Rise Time at TP2 (10% to 90%)		—	—	—	200	ps	—
HDMI Tx (TMDS Source)							
TDR Rise Time at TP1 (10% to 90%)		—	—	—	200	ps	—
Rise/Fall Time: Data		—	42.5	—	—	ps	—
Rise/Fall Time: Clock		—	75	—	—	ps	—
Intra-pair skew: Max		—	—	—	$0.15T_{\text{bit}}$	—	—
Inter-pair skew: Max		—	—	—	$0.2T_{\text{character}}$	—	—
Maximum Differential Voltage V_{high}		—	—	—	780	mV	—
Minimum Differential Voltage V_{low}		—	-780	—	—	mV	—
Clock Duty Cycle		—	40	—	60	%	—
TMDS Clock Rate		—	25	—	294	MHz	—
Clock Period: Bit Period		—	—	40	—	—	—

Latency						
Genlock	—	—	—	120	μs	—
Genlock Scaler & Genlock Wall	—	—	1/4	—	frame	—
Fast Switch & Fast Switch Wall	—	1	—	2	frame	—
Video Wall	—	1	—	2	frame	—
Multiview	—	1	—	2	frame	—
DDR3 (1.35V)						
Clock Period (DLL disable)	—	8	—	7800	ns	—
Clock absolute high pulse width	—	0.43	—	—	tCK	—
Clock absolute low pulse width	—	0.43	—	—	tCK	—
Clock period jitter (DLL locked)	—	-60	—	60	ps	—
Data setup time to DQS	—	25	—	—	ps	—
Data hold time from DQS	—	55	—	—	ps	—
DDR3 (1.5V)						
Clock Period (DLL disable)	—	8	—	7800	ns	—
Clock absolute high pulse width	—	0.47	—	0.53	tCK	—
Clock absolute low pulse width	—	0.47	—	0.53	tCK	—
Clock period jitter (DLL locked)	—	-70	—	70	ps	—
Data setup time to DQS	—	25	—	—	ps	—
Data hold time from DQS	—	45	—	—	ps	—
I ² C						
SCL clock frequency	—	0	—	400	kHz	—
Rise time	Both SDA & SCL	20	—	300	ns	—
Fall time	Both SDA & SCL	20 x (V _{DD} / 5.5)	—	—	ns	—

MDIO (10Gb/s)						
Input Capacitance	—	—	—	10	pF	—
Bus loading	—	—	—	470	pF	—
Driver: AC common mode output voltage	30Hz to 40kHz	—	—	2.5	V	—
Driver: DC common mode output voltage	—	—	—	5.5	V	—
Receiver: AC differential input impedance	—	—	78Ω + 50μH	—	—	—
Receiver: AC common mode range	30Hz to 40kHz	0	—	3	V	—
Receiver: Total common mode range	—	0	—	5.5	V	—
XFI (10Gb/s)						
Tx Total Jitter	—	0	—	610	mUI	—
Tx Eye Width	—	59	—	—	ps	—
Tx Eye Height	—	60	—	410	mV	—
Tx Jitter Generation	—	0	—	7	mUI _{RMS}	—
Tx AC Common Mode Voltage	—	0	—	25	mV _{RMS}	—
Tx Return Loss	SDD22	—	-20	—	-1	dB
Tx Return Loss	SCC22	—	-3	—	-3	dB
Rx Jitter Tolerance	0.04MHz	6.8	—	—	UI	—
	4MHz	0.232	—	—	UI	—
	8MHz	0.212	—	—	UI	—
	27.2MHz	0.062	—	—	UI	—
	80MHz	0.062	—	—	UI	—
Rx Return Loss	SDD11	—	-20	—	-1	dB
Rx Return Loss	SCC11	—	-3	—	-3	dB

I ² S (Audio)							
Tx delay time	—	—	—	0.8T _r	—	—	—
Tx Hold time	—	0	—	—	—	—	—
Rx Setup time	—	—	—	0.2T _r	—	—	—
Rx Hold time	—	—	—	0	—	—	—
Tx Clock Period	—	T _{tr}	—	—	—	—	—
Rx Clock Period	—	T _r	—	—	—	—	—
Master mode Clock High (Tx & Rx)	—	0.35T _{tr}	—	—	—	—	—
Master mode Clock Low (Tx & Rx)	—	0.35T _{tr}	—	—	—	—	—
Slave mode Clock High (Tx & Rx)	—	—	—	0.35T _{tr}	—	—	—
Slave mode Clock Low (Tx & Rx)	—	—	—	0.35T _{tr}	—	—	—
Slave mode Rise time (Tx)	—	0.15T _{tr}	—	—	—	—	—
RGMII							
Clock frequency / accuracy	—	125 - 100 ppm	—	125 + 100 ppm	MHz	—	—
Data to Clock output Skew	T _{skewT}	—500	0	500	ps	—	—
Data to Clock output Skew	T _{skewR}	—	1	1.8	2.6	ns	—
Data to Clock output setup	T _{setupT}	—	1.2	2	—	ns	—
Clock to Data output hold	T _{holdT}	—	1.2	2	—	ns	—
Data to Clock input setup Setup	T _{setupR}	—	1	2	—	ns	—
Data to Clock input hold Setup	T _{holdR}	—	1	2	—	ns	—
Clock Cycle Duration	T _{cyc}	—	7.2	8	8.8	ns	—
Duty Cycle for Gigabit	Duty _G	—	45	50	55	%	—
Rise / Fall Time (20% to 80%)	T _r / T _f	—	—	—	0.75	ns	—

3 Detailed Description

3.1 BlueRiver Architecture

Figure 3-1 below outlines a fully loaded BlueRiver design demonstrating the main hardware components and interfaces that are included in BlueRiver designs.

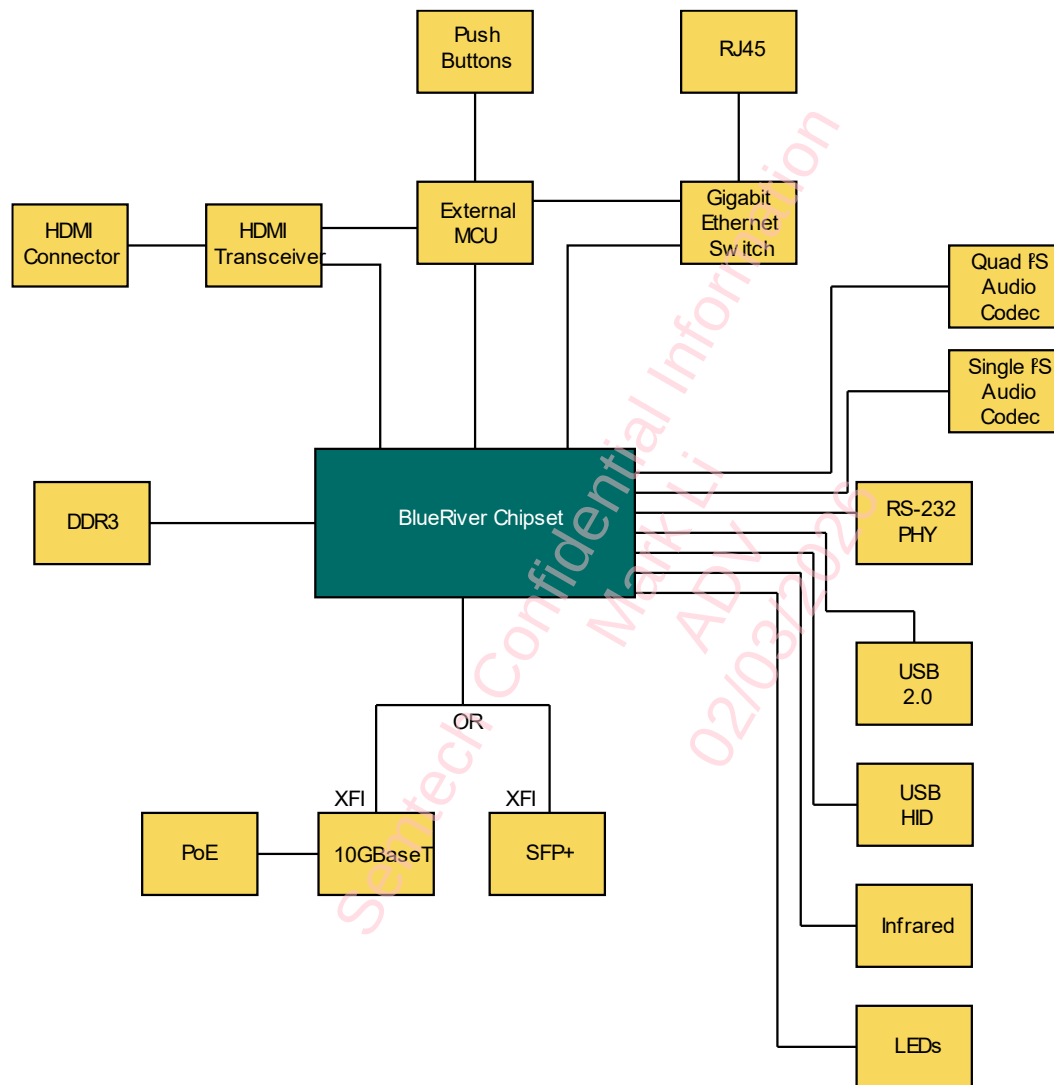


Figure 3-1: BlueRiver Components and Interfaces

3.2 BlueRiver AV Processor (AVP2000 and AVP2000T)

The BlueRiver AV processor is a fully in-line engine for high-fidelity processing of uncompressed audio and video signals. It integrates numerous functions into a single, in-line, zero-frame latency signal processing subsystem. Available on the BlueRiver AVP2000/AVP2000T devices.

These functions include:

- Broadcast quality scaling
- Multi-source video compositing
- Video cropping
- Colour space conversion
- Frame rate conversion

The BlueRiver AV processor is fully embedded into and tightly coupled with the BlueRiver chipset's signal extension and switching capabilities delivering a single-chip solution for end-to-end signal distribution and processing.

3.2.1 Broadcast Quality Scaling

Optimized for video, graphics and text quality, the BlueRiver AV processor supports both up-scaling and downscaling for resolutions up to True 4K60/4:4:4.

This broadcast quality scaling engine is architected with fully programmable horizontal and vertical taps and includes pre and post processing algorithms to reduce ringing and aliasing artefacts while preserving the sharpness of edges.

BlueRiver scaling engine operates in-line with other functional blocks of the chipset, such as HDMI encoding/decoding, AES 128-bit encryption, extension (10G streaming) and lightweight compression (if required), adding only 16-lines of latency to the video transmission (maximum of 0.12ms of latency at 4K60). A wide variety of input formats including RGB and YUV (444/422/420) are supported along with 8-bit, 10-bit and 12-bit colour depths. Output of BlueRiver scaler is always RGB 8-bit.

3.2.2 Scaling Applications

Support for in-line scaling enables BlueRiver extension and switching chipsets to deliver commonly used Pro AV applications. This section provides some examples.

3.2.2.1 Display Adaptive Scaling 1 × n Signal Distribution

When a source is being transmitted to multiple displays of varying resolutions, the scaling engine in each receiver can automatically upscale or downscale the source to the native resolution of the attached display, ensuring that each display receives the source in its optimal image quality.

Without this capability, the source must be set to the resolution of the lowest resolution display forcing all higher resolution displays to show a lower quality down-scaled image.

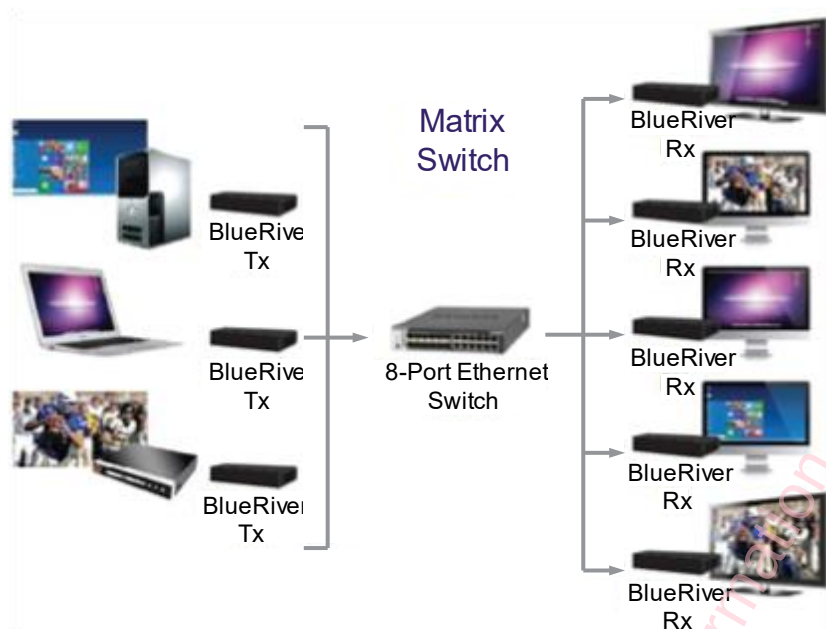


Figure 3-2: Adaptive Scaling 1 × n Signal Distribution

3.2.2.2 True Seamless Switching in n × m Signal Distribution

In a Matrix Switching environment, switching a display from one source to another typically results in a black screen, frozen frames or other artefacts.

With the BlueRiver AV processor's scaling engine integrated into an AV-over-IP platform, truly seamless switching is achieved where the switching delay is only 1-2 frames and without any artefacts, irrespective of the resolution of the switched source.

3.2.2.3 Multiview, PIP, and PAP

The scaling engine works together with the multi-source video compositing capabilities of the BlueRiver AV processor to deliver multi-view, picture-in-picture (PIP) and picture and picture capabilities (PAP) within the AV signal distribution system, without the need for any additional equipment.

3.2.2.4 Video Wall Processing

When used with the video-cropping and bezel correction capabilities of the BlueRiver AV processor, the scaling engine enables full video-wall capabilities without the use of additional equipment.

3.2.2.5 Video Cropping and Bezel Correction

The BlueRiver AV processor allows each BlueRiver receiver to automatically crop an incoming video and render only a portion of the image as part of a video wall.

Configurable bezel correction support compensates for the thickness of the bezel to virtually hide the pixels that need to be obscured by the display bezel. Combined with the scaling engine, this capability delivers video walls of any size and eliminates the need for additional equipment.

3.2.2.6 BlueRiver Compression

BlueRiver extension and switching chipsets transmit uncompressed, pixel-for-pixel video signals extended over long distances and switched through IP switches with zero-frame latency.

In a few cases, it is required to apply lightweight compression to enable a video resolution to be transmitted across a limited bandwidth network or enable longer distance transmission for the same resolution over lower data rate NBaseT connectivity.

For such situations, a unique compression algorithm is employed with a compression ratio of just 1.4:1. When enabled, this lightweight compression ratio results in visually indistinguishable image quality.

BlueRiver Compression is a high-quality compression technology designed for the very specific task of compressing 12G 4K signals, such as 4K60 8-bit 4:4:4 or 4K60 10-bit 4:2:2, to fit within a 10GbE infrastructure using approximately 9G of bandwidth, leaving 1G for general purpose network data.

Latency is limited to five line periods (under 8 microseconds for 4K60 video). Video quality that is indistinguishable from native video and with imperceptible latency.

Lightweight video compression is also automatically applied to video formats to reduce link bandwidth for carriage over both 2.5GBaseT and 5GBaseT. For example, 1080p60 4:4:4 over 2.5GBaseT or 4K30 4:4:4 over 5GBaseT.

3.2.2.7 Multi-Source Video Compositing

BlueRiver AV processor's multi-source video compositing capability allows each BlueRiver receiver to combine multiple video sources onto a single display enabling a variety of applications such as Multiview, PIP and PAP.

- The multi-source video compositing capability is tightly coupled with the scaling engine which is used to scale the resolution of each source to the target window on the display canvas
- The 4K60 capable compositing engine can tolerate many asynchronous inputs and can render without tearing or other visual artefacts
- Multiple pre-sets are available and layouts can be updated in real-time

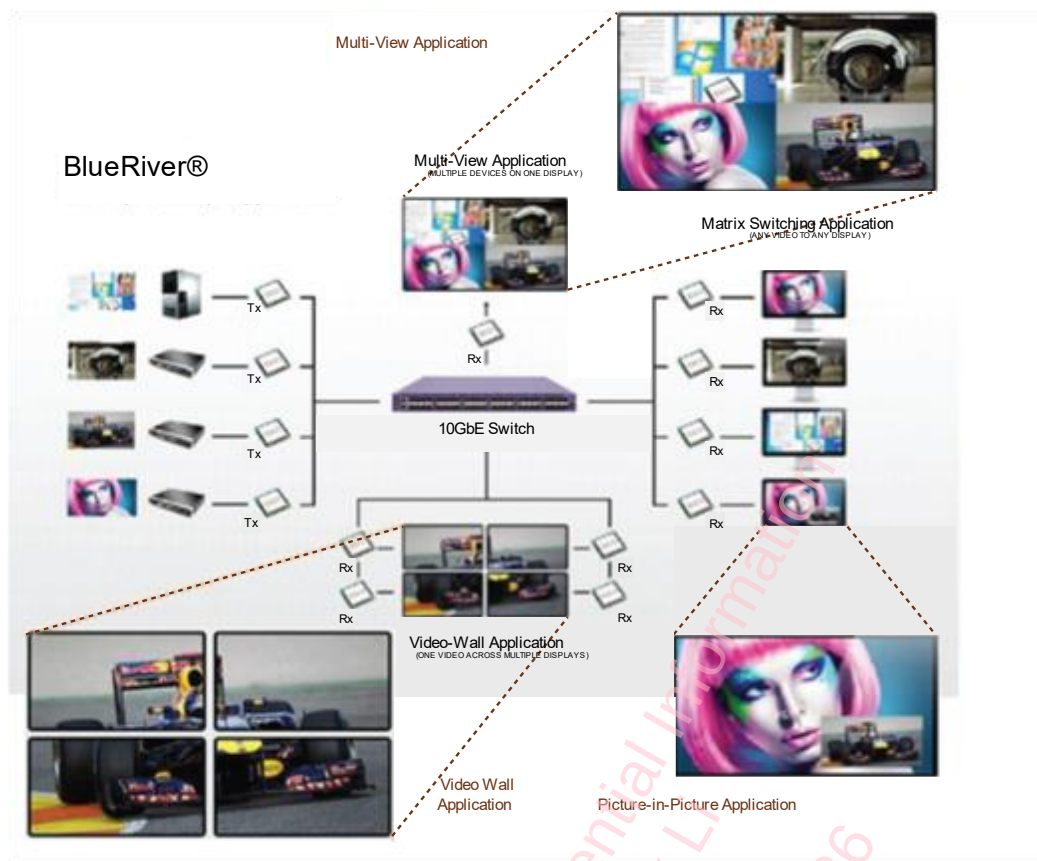


Figure 3-3: Multiple Source Compositing (Video Wall / Multiview)

3.2.2.8 Colour Space Conversion

Often pro AV applications require the flexibility of supporting many types of input formats, such as RGB, YUV 4:2:0 and YUV 4:2:2. Mismatches between the format being output by the source and the formats supported by the display are common, resulting in interoperability issues.

The BlueRiver AV processor integrates an in-line colour space converter, which automatically converts between mismatching colour spaces between the source and the display.

Colour Space Conversion Definition:

Colour space conversion is the translation of the representation of a colour from one basis to another. This typically occurs in the context of converting an image that is represented in one colour space to another colour space, the goal being to make the translated image look as similar as possible to the original.

- When set in the Genlock mode, the receiver outputs the HDMI video in exactly the same format as it was received from the transmitter connected to the source.
- For the AVP2000 and AVP2000T, when the receiver is set to any other processing mode, the receiver output is always in 8-bit RGB colour space.
- The BlueRiver AVP2000 and AVP2000T also offer the additional capabilities to both Chroma re-sample YUV 4:2:2 and YUV 4:2:0 sources to RGB, as well as convert both 10-bit and 12-bit sources into 8-bit.

3.2.2.9 Frame-Rate Conversion

The BlueRiver AV processor includes full arbitrary frame rate conversion, enabling true cross-conversion from any resolution and frame rate to another. For instance:

- 4K30 can be scaled down to 1080p60
- 1080p24 can be scaled up to 4K60
- Content sourced at 59.97Hz can be used along with 60Hz without special considerations

Other AV processing capabilities of the BlueRiver AV processor, such as scaling and true seamless switching also benefit from frame rate conversion. For example, a monitor does not need to re-sync during an input switch, regardless of the associated resolution or refresh rate.

Note:

The AVP2000T does not support frame rate division on the HDMI input (transmitter path). Please see [Section 3.2.5](#) for more details.

3.2.3 Audio Capabilities

Professional AV applications often have audio requirements as complex as their video requirements. The BlueRiver AV processor provides highly-flexible capabilities to manage these requirements.

3.2.3.1 Stereo Down-Mixing

When configured as a Tx, all BlueRiver devices provide down-mixing of multi channel LPCM audio to stereo, allowing stereo only receive end-points to support the audio from multichannel sources.

3.2.3.2 Audio Embedding and De-Embedding

BlueRiver devices provide the following audio capabilities:

- Independent routing of audio from video.
- BlueRiver transmitter can de-embed audio from HDMI.
- BlueRiver receiver can re-embed audio into HDMI.
- BlueRiver transmitter can down-mix LPCM audio.
- Extraction and insertion of audio using I²S interface.

To enable advanced audio processing, the BlueRiver transmitter allows audio to be extracted from the HDMI source to be output via I²S to other on-board modules, such as DSPs, DACs or other audio subsystems that are included in the hardware design.

After processing, this audio can be reinserted and re-embedded with video and sent via HDMI stream or BlueRiver AV network feed. Analog audio may also be embedded by connecting an ADC to the I²S input port on the BlueRiver device.

3.2.4 Adaptive Clock Recovery (ACR)

Semtech's Adaptive Clock Recovery (ACR) is an algorithm that allows the BlueRiver receiver devices to time-lock HDMI displays to HDMI sources over a network.

Using off-the-shelf network switches for high-quality AV signal distribution present an inherent challenge. Synchronous and artefact-free low latency transmission of video is a requirement for the majority of Pro AV applications. In addition, genlocking is often a requirement for video-wall and multiview applications.

The challenges encountered include:

- Network switches which are asynchronous and cannot transport synchronous video without requiring a frame-buffer and re-clocking at the receiving endpoint
- The use of frame buffers and re-clocking results in a 1-2 frame latency, frame drops and frame repeats. In addition, there is a lack of genlocking, making such systems unsuitable for many Pro AV and broadcast applications

Semtech developed the ACR technology to guarantee synchronous data delivery across asynchronous networks. ACR has been validated by Semtech for up to three switch hops. ACR is used by receivers operating in genlock, genlock scaling or synchronized video wall modes.

ACR works with all 10GbE network switches given that the switch is a standard Layer 2 or Layer 3 Ethernet switch.

3.2.5 AV Transceiver Mode (AVP2000T Only)

The AVP2000T supports simultaneous HDMI Receive and Transmit functionality (Transceiver mode). In this mode, both the TMDS input and output interfaces are active, allowing AV signals over HDMI to be transmitted and received across the 10G Ethernet link using a single device. Since the AVP2000T has one DDR interface with on-chip memory controller, the frame buffer is only available on the HDMI output (receive path). Consequently, frame rate division on the HDMI input (transmitter path) is not supported. However, the device has a scaler on both the transmit and receive path, enabling scaling in both directions simultaneously. Figure 3-4 shows the detailed video and audio signal paths through the AVP2000T in Transceiver mode.

Transceiver – Video/Audio

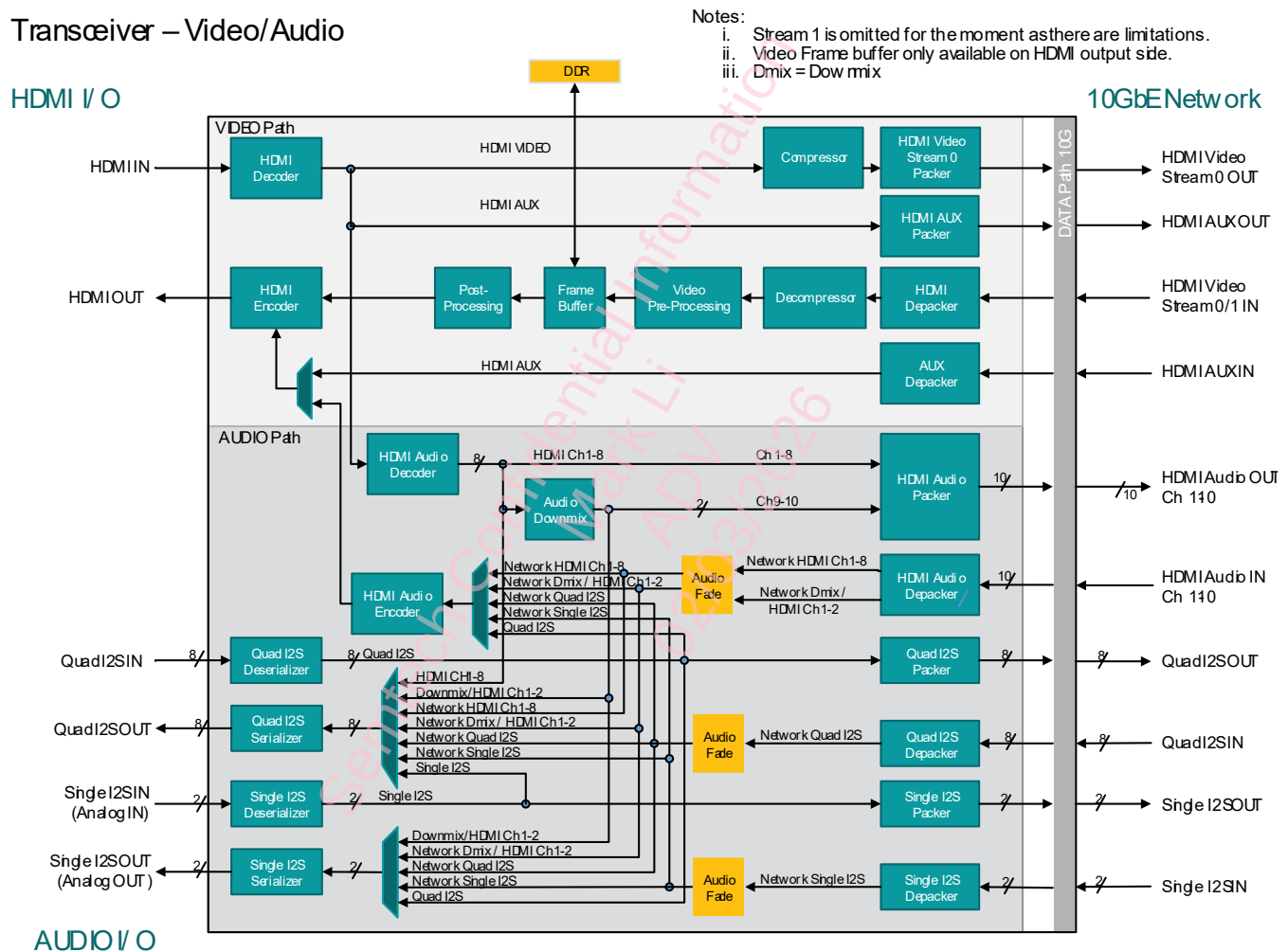


Figure 3-4: Transceiver Diagram

3.2.5.1 AV Local Loop Back

The AV local loop back functionality meets two requirements. An AVP2000T transceiver can output all available remote BlueRiver streams as well as its own local input stream. However, the local loop back function is not supported in Multiview. An AVP2000T transceiver does not support local loop back stream 0 and stream 1 of its HDMI input path in Multiview.

The AV local loop back feature allows the HDMI input to the AVP2000T to be forwarded to the HDMI output of the same device.

Note:

When operating in loop back mode, all transceiver input streams are available to be forwarded to other BlueRiver devices.

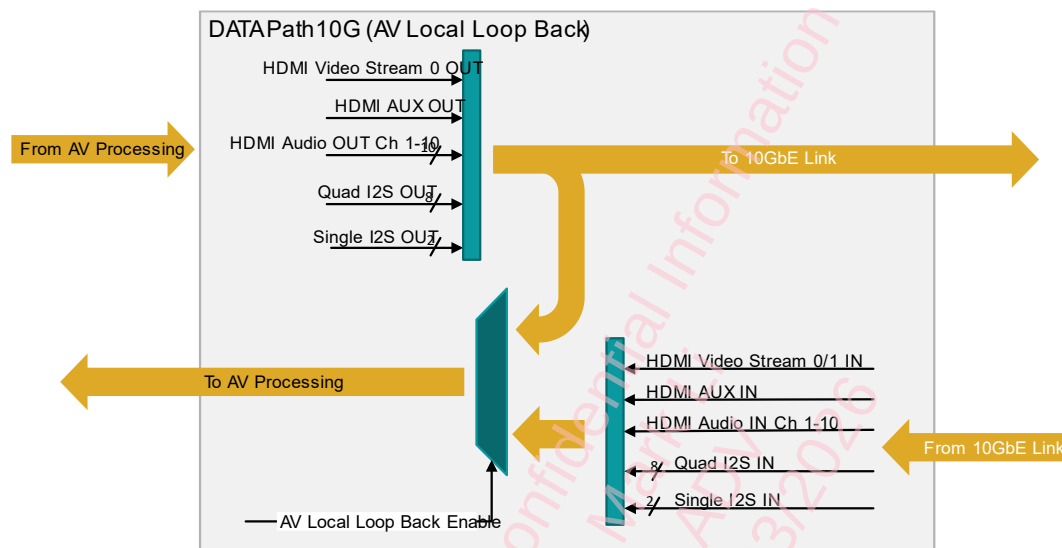


Figure 3-5: AV Local Loop Back Internal Signal Paths

The local loop back configuration is implicit; the Semtech API command set identifies a loop back condition using a “join” command. Details on configuring the local loop back is covered in the SDVoE Developers API Reference Guide (PDS-062489).

If a stream is looped back locally, each stream type is configured independently.

Video loop back operation supports:

- Genlock
- Audio loop back
- For video processing modes the following applies:
 - Fast Switch
 - Genlock scaling
 - Video Wall mode — Genlock
 - Video Wall mode — Fast Switch

It is possible to loop back the native stream (stream 0).

- DDR memory is required to be implemented in the hardware design.

Table 3-1 below describes some limitations in the RGMII interfaces in transceiver mode when the HDMI output is subscribed to its own HDMI input (loop back mode). Such restrictions do not apply when the transceiver is not in loop back mode. Alternatively, customers can avoid these limitations by doing their loop back externally to the BlueRiver chipsets and using the functionality already present in their HDMI front end chip.

Table 3-1: Limitations on the RGMII Interfaces in Loop Back Mode.

HDMI Video Input Format	RGMII — UEX	RGMII — 1G
4K50, 4K60 or UHD	Not Supported	Not Supported
4K30 or 4K24	Not Supported	Max link speed is 100Mb/s
4K30 or 4K24	Supported	Not Supported
1080p	Supported	Supported

3.3 BlueRiver System Components

3.3.1 External Microcontroller

When planning the hardware design of BlueRiver transmitters or receivers, an external microcontroller is required. The external microcontroller unit (MCU) is used to control external subsystem components to the core functionality of the BlueRiver chipset.

It adds flexibility and intelligence to the hardware design, allowing the BlueRiver chipset to seamlessly interact with external systems, such as:

- HDMI Transceiver
- 1GbE Ethernet
- Custom subsystems, such as audio processing subsystems or additional video interfaces such as DisplayPort, VGA, or SDI.

Note:

For a list of components and interfaces managed directly by the BlueRiver chipset refer to Hardware Interfaces.

3.3.2 Subsystems Controlled by the BlueRiver Chipset

The following subsystems are controlled exclusively by the BlueRiver chipsets:

- Clock generator
 - **Reminder!** Applies to receivers only. Transmitters do not require a clock generator
- 10GBaseT PHY
- Semtech UEX USB 2.0 extender chip

These components and interfaces are covered later in this document.

3.3.3 HDMI Transceiver

The HDMI Transceiver is a core component to any BlueRiver design. It converts the 3.3VDC coupled TMDS interface to AC-coupled interface supported by BlueRiver chipsets. It must also contain a valid HDCP key.

This HDMI transceiver is configured as an:

- Input on transmitter devices
- Output on receiver devices
- Input and output on transceiver devices

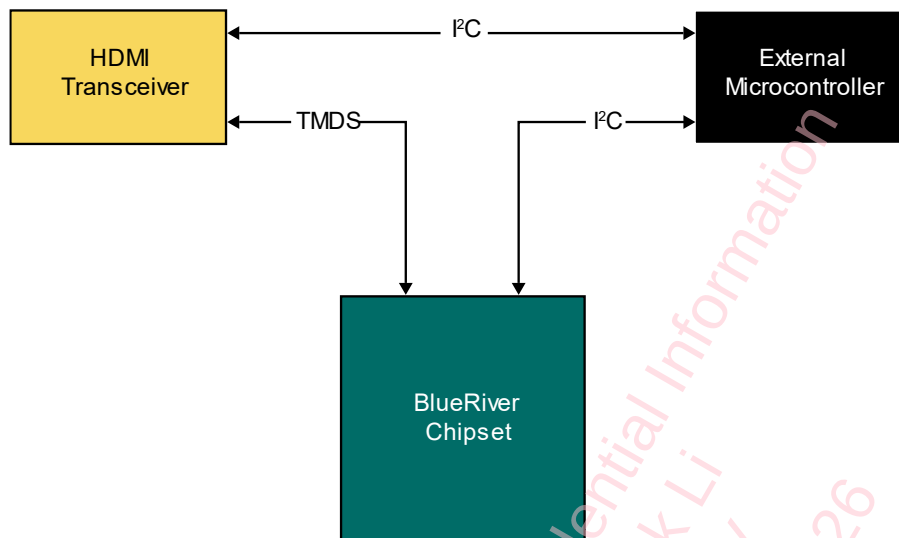


Figure 3-6: HDMI Connection with BlueRiver Chipset and External Microcontroller

3.3.4 1GbE Switch IC

To allow for SDVoE API support as well as for DHCP IP management of a BlueRiver device, an optional 1GbE Ethernet switch may be implemented into the hardware design.

The API is required in implementing advanced features, such as frame rate division or managing a video wall.

The 1G interface can implement an internal delay inside the transmitter as well as inside the receiver.

Note: The RGMII interface has a maximum 4KB packet buffer size. If the device receives data via the RGMII that exceeds this buffer size, this could potentially result in packet loss for 1G traffic.

3.3.4.1 Delay on RGMII Tx

This scenario represents the case where the internal delay is enabled on the Tx side. The delay needs to be implemented on the Tx side in the third-party switch IC as well.

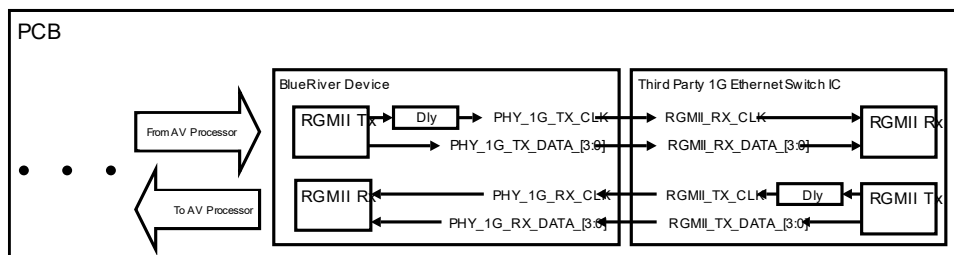


Figure 3-7: RGMII Tx Delay Block Diagram

3.3.4.2 Delay on RGMII Rx

This scenario represents the case where the internal delay is enabled on the Rx side. The delay needs to be implemented on the Rx side in the third-party switch IC as well.

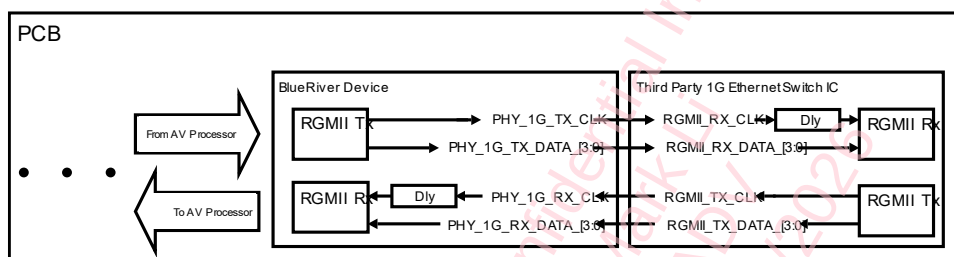


Figure 3-8: Delay on RGMII Rx

3.3.5 DDR3 Memory (AVP2000 & AVP2000T Only)

The DDR3 memory is used by the AVP2000T & AVP2000 device's AV processing engine, referred to as the AV Processor.

The role of the DDR3 memory is to support advanced processing features, including a video frame buffer and video frame rate conversion.

3.3.6 Clock Generator

Silicon Labs SI5341B clock is used by the BlueRiver chipset to regenerate the TMDS clock.

All receiver designs require the Silicon Labs SI5341B clock. The clock generator is not required for transmitter designs.

Semtech have qualified other clock generator part numbers from Silicon Labs, as well as parts from alternate manufacturers such as SiTime and Renesas. For information on alternate clock generator devices that can be used, please see the Hardware Design Guide (PDS-062051).

3.3.7 BlueRiver SPI Flash

The SPI Flash stores the BlueRiver firmware and all non-volatile device settings.

The flash stores two firmware images:

Primary: Default firmware that loads during device boot-up.

Golden image: Fallback firmware.

In the event that a device fails to boot successfully from the primary firmware, it initializes using the golden image firmware. Factory configuration settings and firmware version are applied.

3.3.8 LED Control

Semtech BlueRiver chipsets support a series of LEDs that indicate status and activity, as follows:

- Status
- Video In
- Video out
- 10G Tx Activity
- 10G Rx Activity
- USB 2.0 Present
- USB 2.0 Activity
- USB HID Activity

Refer to Table 1-1: Ball Descriptions for details.

3.4 Hardware Interfaces

3.4.1 I²C Interface and Registers

For the I²C interface protocol and register map descriptions, please refer to the Register Map User Guide (PDS-062203).

3.4.2 USB 2.0 Interface

BlueRiver hardware designs include a USB 2.0 interface that is reserved for use with the Semtech UEX chip to enable USB 2.0 extension functionality for USB devices, such as keyboard, mouse, USB drive and other USB devices.

The USB 2.0 data is transported over RGMII, which is configured to work in 1GbE mode with a 125MHz clock.

3.4.3 USB HID Interface

In addition to support for USB 2.0 connectivity via the Semtech UEX chipset, the BlueRiver devices also supports direct connectivity of USB HID devices for Keyboard, Mouse and Touch Panel support. This capability is provided in addition to Semtech UEX USB 2.0 connectivity, allowing the USB 2.0 and USB HID interfaces to function independently.

The BlueRiver integrated USB HID functionality requires connection to an external USB ULPI PHY device. Semtech recommends USB3318C from Microchip Technology.

The BlueRiver device can be configured in either LOCAL or REMOTE mode. When in LOCAL mode, the device must be connected to an external USB host. When in REMOTE mode, the device must be connected to an external USB hub and supports up to three USB devices. Also, the USB HID REMOTE supports a maximum of 15 endpoints, spread over the three HID devices, as illustrated in Figure 3-9.

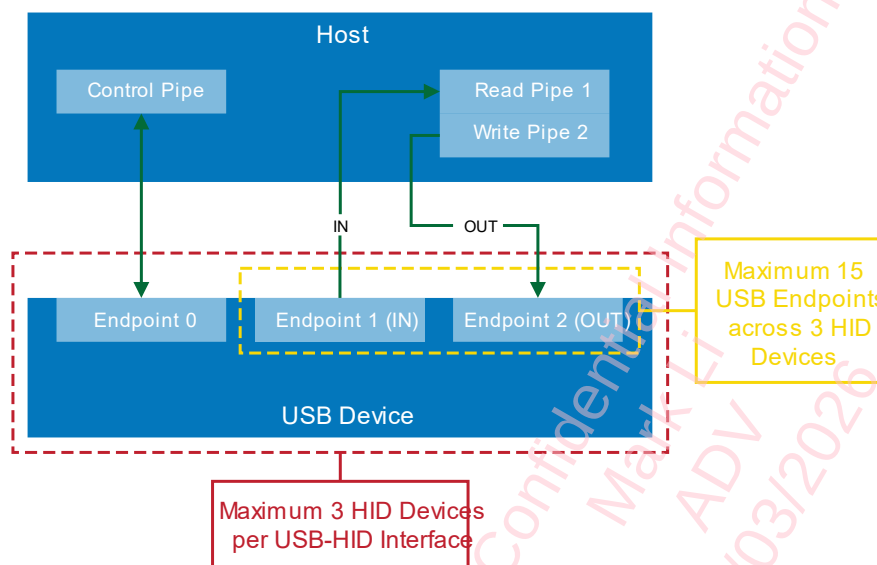


Figure 3-9: USB HID Devices and Endpoints

It is possible to connect a LOCAL and REMOTE device in point-to-point mode. USB devices will be automatically paired when in point-to-point mode.

When configured as LOCAL, the USB HID interface always operates at Full Speed (12Mb/s). In REMOTE mode, USB devices will connect as High Speed, but the actual payload bandwidth will be limited to Full Speed.

When configured in REMOTE mode, the USB HID interface can operate in Low, High or Full Speed. However, if a USB 1.1 hub is used, only Full Speed devices connected to this hub will work. A USB 2.0 hub is recommended to operate with both Low and Full Speed devices. Older USB devices with a built-in USB hub may need to be connected through a USB 2.0 hub to support all functionality.

USB HID streams can only be connected between endpoints over the 10GbE network interface. To send USB HID streams to the 1GbE port on the device, the stream must be routed to an IP address on the 1GbE network. See PDS-062253 for more details.

3.4.3.1 Terminology

- LOCAL mode is the mode where the BlueRiver chipset USB port is connected to an external USB Host. LOCAL refers to the local extender.
- REMOTE mode is the mode where the BlueRiver chipset USB port is connected to the USB device either directly or through an external USB HUB. REMOTE refers to the remote extender.

3.4.3.2 LOCAL/REMOTE Modes

The LOCAL/REMOTE modes are configurable.

These modes are independent for the receiver (RX) and transmitter (TX) devices. Both the receiver (RX) and transmitter (TX) devices may be configurable to operate in either the LOCAL or REMOTE mode. The transceiver device can also operate in both modes.

Tip:

The mode hardware settings are managed through the device firmware when customer products are created using the BlueRiver Production Assistant tool. Refer to the BlueRiver Production Assistant User Guide (PDS-061773) for more information on managing device hardware settings.

3.4.3.3 Local Extender (LOCAL)

The local extender (LOCAL) device is connected to an external host.

Only one host can be connected on the physical USB through the USB PHY.

Note:

The LOCAL device can present one host device, like PC or media system. It does not connect to a mouse or keyboard (that is the remote).

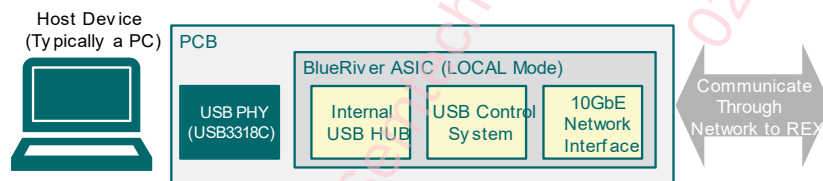


Figure 3-10: SUB Local Extender (LOCAL) Mode

3.4.3.4 BlueRiver Remote Extender (REMOTE) Options

With the use of an external USB HUB, a remote extender (REMOTE) device can support up to three USB devices.

The BlueRiver chipset USB port connects to an external USB HUB.

This external USB HUB can either be:

1. Situated outside the product PCB, refer below to Figure 3-11: REMOTE USB HUB Port Connected to External USB HUB.
2. Implemented as an USB ULPI PHY on the BlueRiver chipset PCB, refer to Figure 3-12: REMOTE USB HUB Port Connected on Product PCB.

REMOTE USB HUB external to PCB

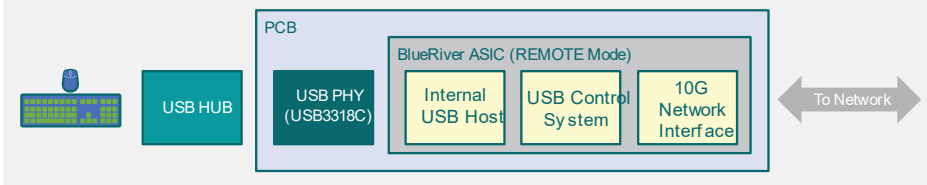


Figure 3-11: REMOTE USB HUB Port Connected to External USB HUB

REMOTE USB HUB implemented as IC on the PCB

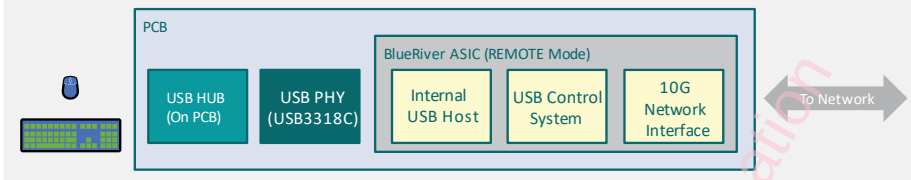


Figure 3-12: REMOTE USB HUB Port Connected on Product PCB

Alternatively, the BlueRiver chipsets USB port can be connected directly to a USB device, refer to Figure 3-13: REMOTE USB Port Connected Directly to USB Device.

- In this configuration, only **one** USB device is supported. Limited support is provided for older USB HID devices with a built-in Full Speed hub. In this case, it is recommended to connect the device through an external USB 2.0 hub.

REMOTE directly connected to USB PHY

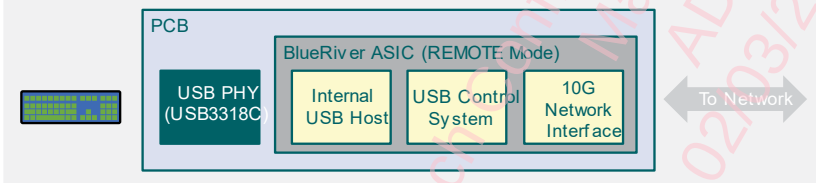


Figure 3-13: REMOTE USB Port Connected Directly to USB Device

3.4.3.5 USB-HID Configuration: Point-to-Point Option

When configured to operate in point-to-point, the USB devices connected to the remote extender (REMOTE) automatically link to the local extender (LOCAL).

When a USB device connects or disconnects, it is automatically detected without user intervention. The state is then forwarded to the LOCAL to notify the host of the state change immediately.

The USB device descriptor presented to the host is a copy of the descriptors of the devices connected on the remote extender (REMOTE).

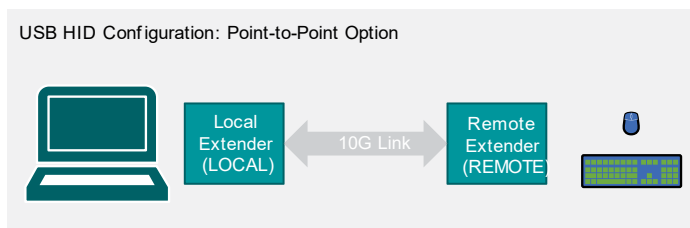


Figure 3-14; USB-HID Configuration—Point-to-Point Option

3.4.3.6 USB-HID Configuration: Switch Option

All local extender (LOCAL) and remote extender (REMOTE) devices communicate through the 10GbE network interface of the BlueRiver chipset.

A local extender (LOCAL) and a remote extender (REMOTE) are paired using SDVoE API commands.

When paired, all devices connected on the remote extender (REMOTE) link directly with the paired local extender (LOCAL).

A device's descriptor is configured through SDVoE API commands.

- When configured, the descriptor presented to the external host are always present, whether the device on the remote extender (REMOTE) is connect or disconnected.
- When not configured, the descriptor presented on the local extender (LOCAL) follows the descriptor of the device connected on the remote extender (REMOTE).
- If the device presented on the remote extender (REMOTE) is disconnected, the local extender (LOCAL) present the device as disconnected to the host.

A remote extender (REMOTE) can be paired to only one local extender (LOCAL).

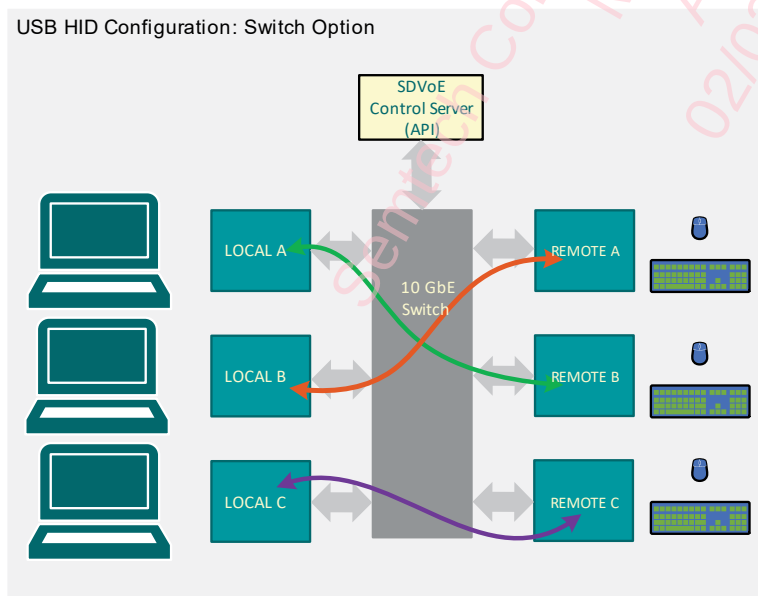


Figure 3-15: USB-HID Configuration—Switch Option

3.4.3.7 USB HID Configuration: KVM Server Option

All local extender (LOCAL) and remote extender (REMOTE) devices communicate through the 10GbE network interface of the BlueRiver chipset.

A local extender (LOCAL) and remote extender (REMOTE) are paired to a KVM server using a SDVoE API command.

The KVM server is responsible for routing all USB HID streams between the local extender (LOCAL) device and remote extender (REMOTE) device.

The protocol used between the local extender (LOCAL)/remote extender (REMOTE) devices and the KVM server is open. System designers must manage streams according to their product requirements.

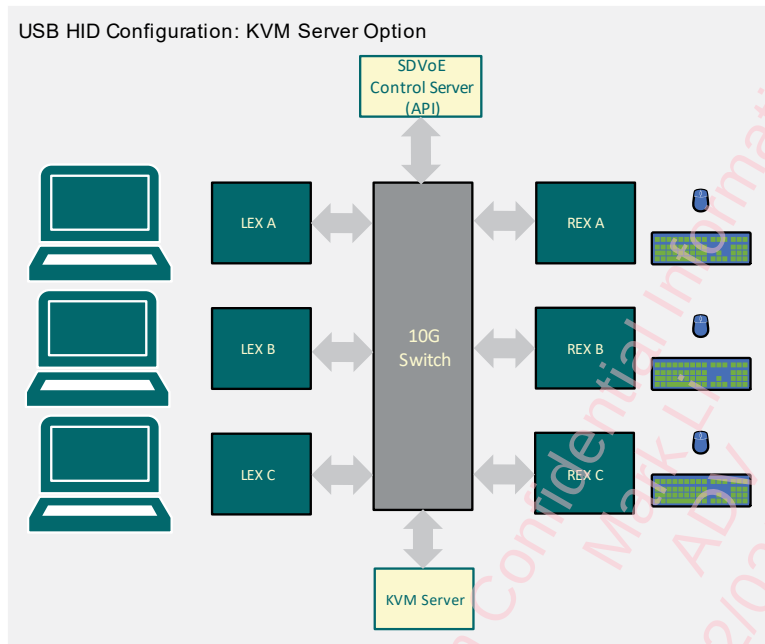


Figure 3-16: USB HID Configuration—KVM Server Option

3.4.4 Quad I²S Interface

All BlueRiver devices include synchronous multichannel I²S interfaces for extracting and injecting HDMI audio or to transport audio between BlueRiver devices.

I²S input consists of quad (4x) synchronized slave inputs, each carrying 2 audio channels for a total of 8 channels.

I²S output consists of quad (4x) synchronized master outputs, each carrying 2 audio channels for a total of 8 channels.

Note:

When auxiliary audio capabilities are being considered, it is advised to implement the quad I²S interface in all new designs.

The quad I²S inputs and outputs support the following:

- Left justified (MSB first) and I²S modes
- 24-bits
- up to 192kHz sampling rate
- Data generated on clock falling edge
- Data sampled on clock rising edge

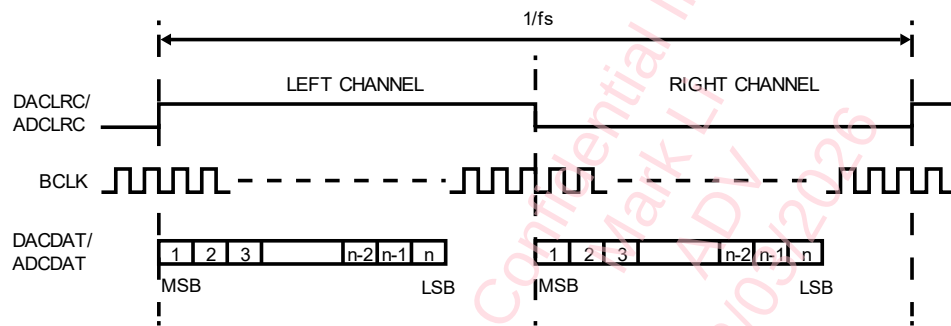


Figure 3-17: Quad I²S Audio Interface Demonstrating Left Justified Mode

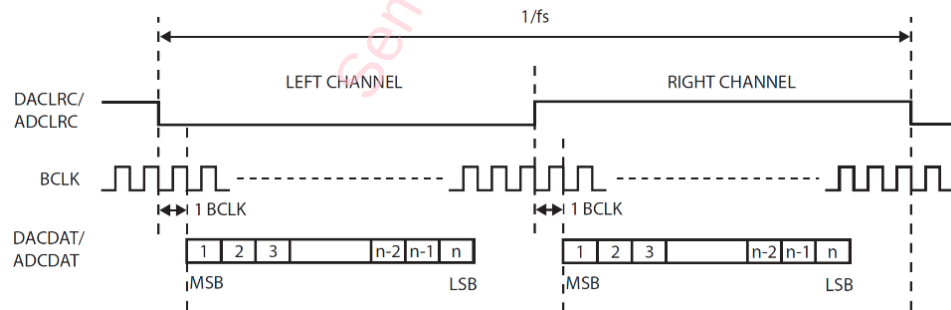


Figure 3-18: Quad I²S Audio Interface Demonstrating I²S Mode

3.4.5 Single I²S Interface

The purpose of this I²S master interface is to transport auxiliary 2-channel audio signals.

This interface can be configured as either an input or output.

When BlueRiver devices are configured in transmitter mode, they can also be used to locally extract 2-channel down-mixed HDMI LPCM audio.

BlueRiver hardware reference designs have an external Cirrus Logic WM87331SEDS codec to the external 2-channel analog audio port. When using the WM8904CFELF, the chip must be managed by an external microcontroller.

When the I²S interface is configured as an input, the audio must be formatted as follows:

- Left justified, MSB first
- LPCM
- 24 bits
- 48kHz
- Data generated on clock falling edge
- Data sampled on clock rising edge

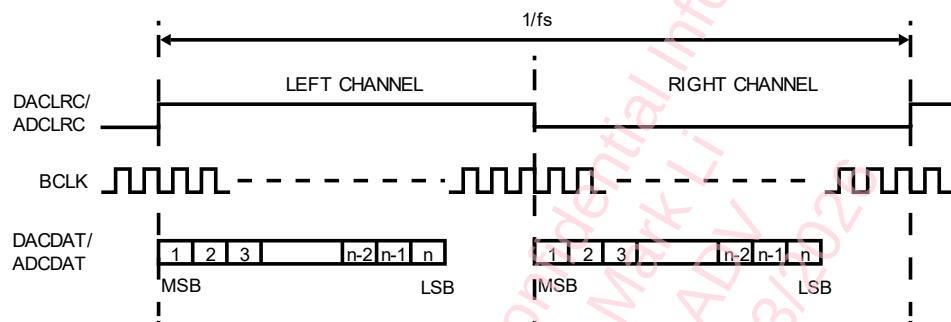


Figure 3-19: Single I²S Audio Interface (Master)

3.4.6 XFI Interface

The XFI interface is used to connect to a discrete external 10GBaseT PHY for 10GbE copper connectivity, or a standard SFP+ optical module for fibre connectivity. The XFI interface has some important characteristics:

- It is a high-speed interface running at 10.3125 Gb/s
- Each transmitter and receiver interface must be routed differentially keeping a differential impedance of 100Ω
- Special care needs to be taken when routing this interface. No vertical interconnect access (VIA) should be used while routing these lines

3.4.7 GPIO Balls

The GPIO balls are available for custom use. By default, all GPIO are tristate (set as inputs).

3.4.8 Infrared

BlueRiver Infrared (IR) I/O operates at 3.3 volts.

It is expected that the BlueRiver infrared output circuit is connected to an infrared emitting device and the BlueRiver infrared input is connected to an infrared receiving device.

The infrared output is driven by the BlueRiver chipset and must be amplified to 5V.

The infrared input is driven by the infrared receiving device and must be dropped to 3.3V, in order to pass the data to the BlueRiver chipset. It is recommended to use a 5V supply for the infrared input.

3.4.9 RS-232

All BlueRiver devices support up to four UART interfaces.

UART1, UART2, UART3 and UART4 may be used to implement other RS-232 custom applications. Baud rates ranging from 2 400 through 115 200 are supported.

Note:

Although BlueRiver hardware schematics list UART ports as 1 through 4, the SDVoE API manages the ports as 0 through 3.

3.4.10 Restore Factory Default Settings

All BlueRiver devices support a factory reset via the *FACTORY* ball. This active-high ball is used to restore factory default settings of the device. In mission mode this ball should be held LOW.

The *FACTORY* ball must be asserted for at least 100ms to ensure proper detection by the internal micro.

The *FACTORY* ball must be de-asserted for at least another 100ms before asserting another factory reset.

3.4.10.1 Push Buttons

The BlueRiver chipset provides a connection that can be connected to the external microcontroller to allow for the chipset to be restored to factory defaults. Refer to Figure 3-20: Example of Controlling Reset Push Button.

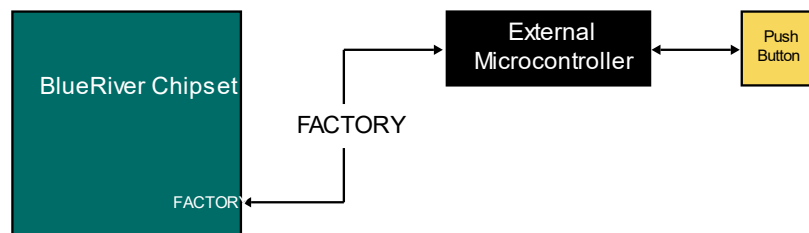


Figure 3-20: Example of Controlling Reset Push Button

3.4.11 Boot Configuration

The BlueRiver device firmware code is stored in external FLASH memory connected via the SPI interface. The firmware is managed through the Semtech Production Assistant tool, which is used to configure HDMI receive and transmit functionality, and which peripherals are enabled, depending on the device variant. Each BlueRiver device also stores bootloader code in on-chip ROM which is required for first time programming and back up in the case of FLASH memory corruption.

During the first time programming, the bootloader provides communication access to program a blank FLASH device. This can be done via three different methods:

1. RS-232
2. Direct UDP connection
3. Through the SDVoE API

Since the bootloader is stored in ROM, the BlueRiver device is still able to run even if the FLASH memory is corrupted. The device firmware is able to run two different programs from FLASH: Primary image and Golden image. During normal device operation, the Main application is always used. If the Primary image fails to load due to, for example, corrupted FLASH memory, the Golden image will be executed.

The Golden image is programmed during initial production, is fully operational code and typically an older version of the Primary image. The main purpose of the Golden code is to enable remote programming of the Primary image in FLASH memory. In the case where both the Primary and Golden image fail to execute, the bootloader provides a backup to maintain device operation until the FLASH can be reprogrammed.

For more details of the bootloader operation, please refer to PDS-062194.

4 Applications Information

For reference designs and detailed application notes, please contact your local Semtech sales and technical support representative.

Semtech Confidential Information
Mark Li
ADV
02/03/2026

UNIT : mm

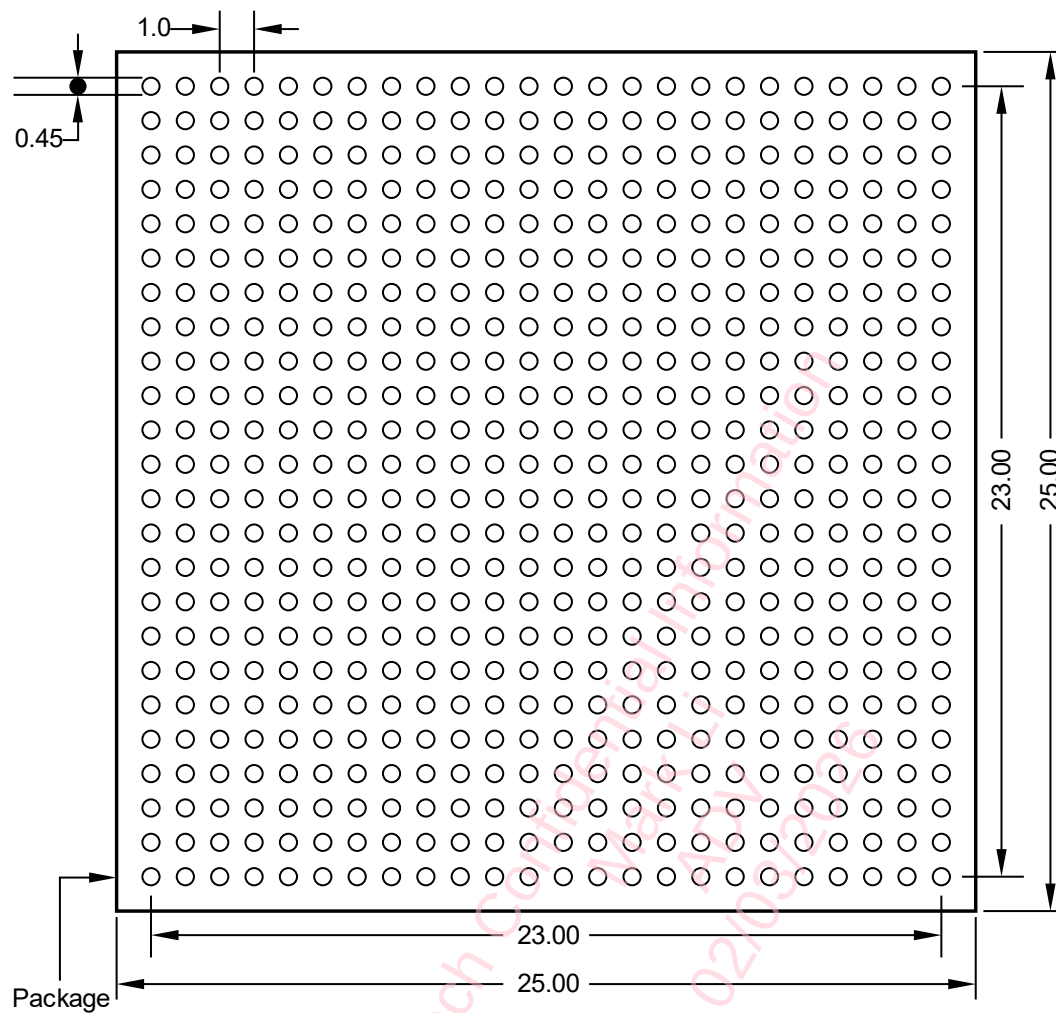
Top View: Dimensions include 25.0, 24.6, 18.6, and 25.0. Features include A1 CORNER, A, B, and a 0.2 (x4) feature.

Side View: Dimensions include 2.83 Max, 0.5 ± 0.1, and 0.2. Features include S, S, and S.

Bottom View: Dimensions include 1.00, 1.00, and 1.00. Features include A1 CORNER, A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, T, U, V, W, Y, and a 576 x Ø0.6 ± 0.1 feature.

AVP1000, AVP2000 & AVP2000T
Data Sheet Rev.9
PDS-061874 March 2025

5.2 Recommended PCB Footprint



NOTE: All dimensions are in millimeters.

Figure 5-2: Recommended PCB Footprint

5.3 Packaging Data

Table 5-1: Packaging Data

Parameter	Value
Package Type	576 ball BGA / 25mm x 25mm / 1.0mm pad pitch
Moisture Sensitivity Level	MSL4
Junction to Air Thermal Resistance, θ_{j-a} (zero airflow)	14.0 °C/W
Junction to Board Thermal Resistance, θ_{j-b}	6.26 °C/W
Junction to Case Thermal Resistance, θ_{j-c}	1.14 °C/W
Junction to Top Characterization Parameter, ψ (Psi)	1.02 °C/W
Pb-free and RoHS compliant	Yes

5.4 Marking Diagram

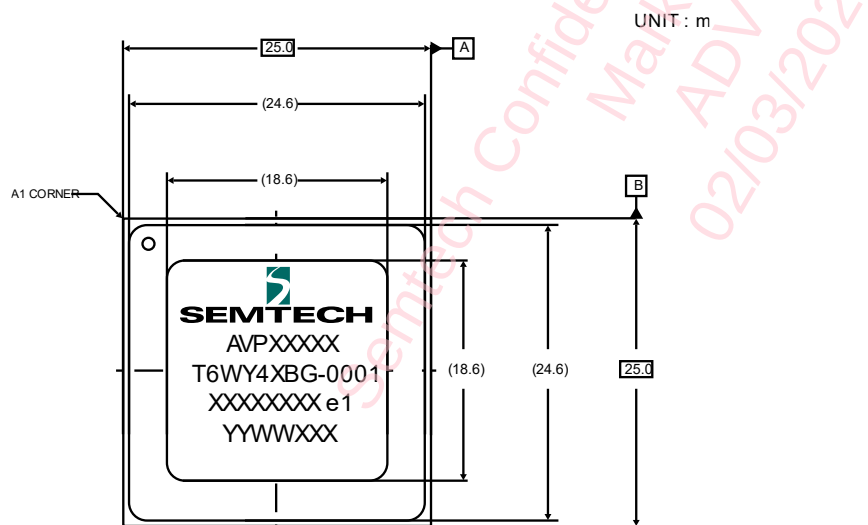


Figure 5-3: Marking Diagram

- AVPXXXXX – Semtech part number.
- T6WY4XBG-0001 – manufacturer part number.
- XXXXXXXX e1 – batch code (5 to 8 characters); e1 indicates Pb-free.
- YYWWXXX – YYWW year and week number of manufacture; XXX manufacturer version identifier:
 - -ES indicates Engineering Sample.
 - HA2 indicates Production Device with Kyocera substrate.
 - HB2 indicates Production Device with KCC substrate.

5.5 Solder Reflow Profile

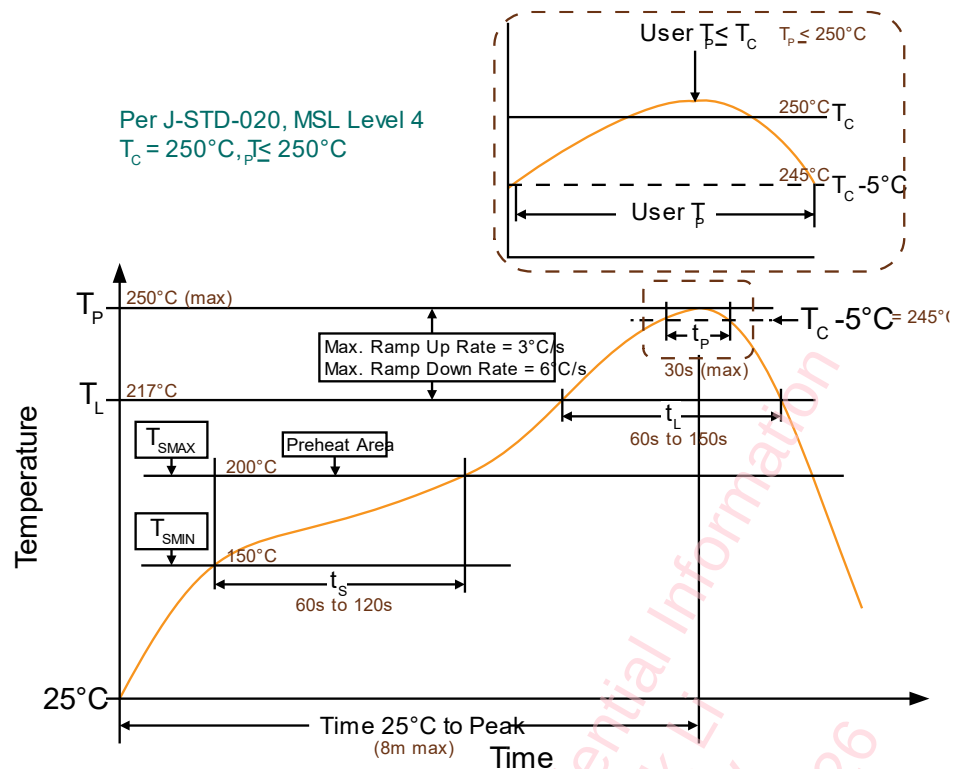


Figure 5-4: Solder Reflow Profile

5.6 Ordering Information

Part Number	Device Package	Shipping Package MOQ
AVP1000-IBE3	BGA576	40 pcs./tray
AVP2000-IBE3	BGA576	40 pcs./tray
AVP2000T-IBE3	BGA576	40 pcs./tray



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